

Lab-02 (CCN)

THURSDAY, JANUARY 29, 2026

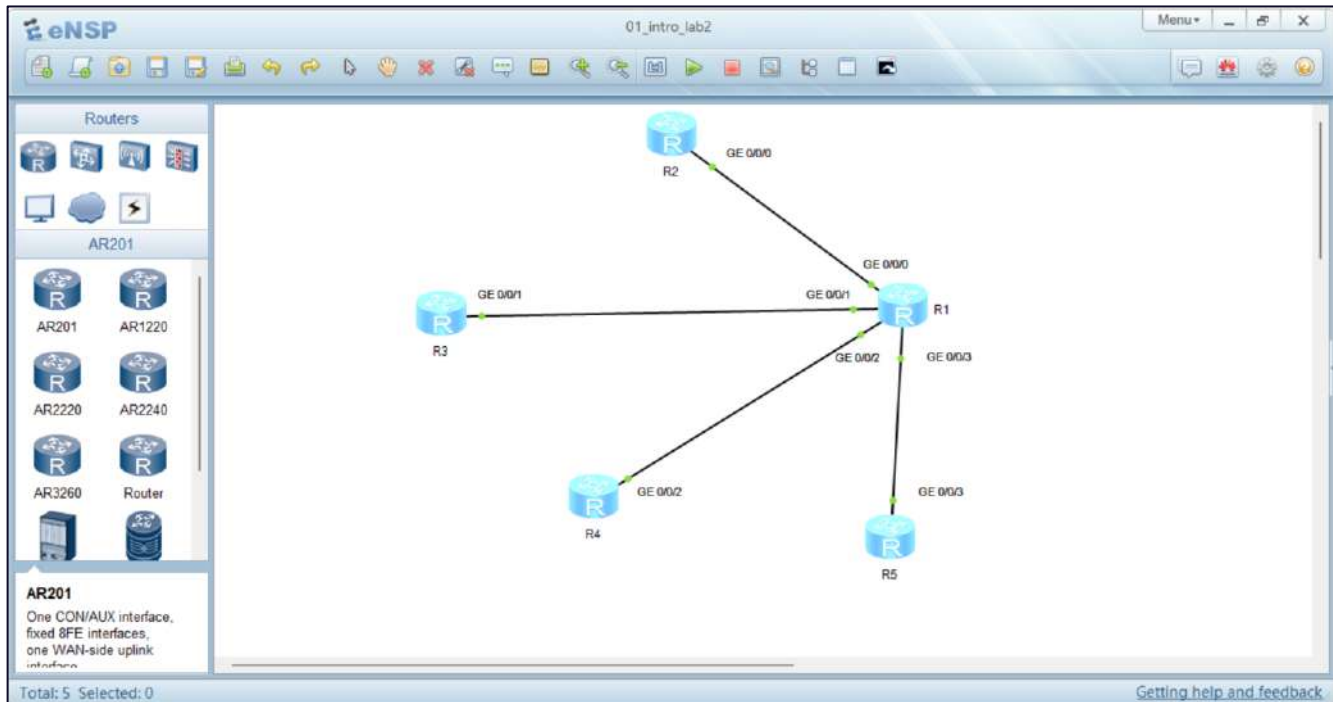
NED University

Name: MUHAMMAD OBAID

Roll Number : CT-23025 (Section A)



IP Assignment for VLSM



Router 1:

```

R1
(s): spoofing
(d): Dampening Suppressed
The number of interface that is UP in Physical is 5
The number of interface that is DOWN in Physical is 6
The number of interface that is UP in Protocol is 5
The number of interface that is DOWN in Protocol is 6

Interface                IP Address/Mask      Physical  Protocol
Ethernet0/0/0             unassigned           down      down
Ethernet0/0/1             unassigned           down      down
GigabitEthernet0/0/0      24.23.5.202/30       up        up
GigabitEthernet0/0/1      24.23.5.206/30       up        up
GigabitEthernet0/0/2      24.23.5.210/30       up        up
GigabitEthernet0/0/3      24.23.5.214/30       up        up
NULL0                    unassigned           up        up(s)
Serial0/0/0               unassigned           down      down
Serial0/0/1               unassigned           down      down
Serial0/0/2               unassigned           down      down
Serial0/0/3               unassigned           down      down
[R1]ping 24.23.5.213
  PING 24.23.5.213: 56 data bytes, press CTRL_C to break
    Reply from 24.23.5.213: bytes=56 Sequence=1 ttl=255 time=40 ms
    Reply from 24.23.5.213: bytes=56 Sequence=2 ttl=255 time=20 ms
    Reply from 24.23.5.213: bytes=56 Sequence=3 ttl=255 time=40 ms
    Reply from 24.23.5.213: bytes=56 Sequence=4 ttl=255 time=30 ms
    Reply from 24.23.5.213: bytes=56 Sequence=5 ttl=255 time=50 ms

  --- 24.23.5.213 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 20/36/50 ms

[R1]quit
<R1>save
The current configuration will be written to the device.
Are you sure to continue?[Y/N]Y
Now saving the current configuration to the slot 17.
Save the configuration successfully.

```

Router 2:

```
R2
*down: administratively down
!down: FIB overload down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
The number of interface that is UP in Physical is 2
The number of interface that is DOWN in Physical is 9
The number of interface that is UP in Protocol is 2
The number of interface that is DOWN in Protocol is 9

Interface                IP Address/Mask    Physical    Protocol
Ethernet0/0/0            unassigned         down       down
Ethernet0/0/1            unassigned         down       down
GigabitEthernet0/0/0     24.23.5.201/30    up         up
GigabitEthernet0/0/1     unassigned         down       down
GigabitEthernet0/0/2     unassigned         down       down
GigabitEthernet0/0/3     unassigned         down       down
NULL0                    unassigned         up         up(s)
Serial0/0/0              unassigned         down       down
Serial0/0/1              unassigned         down       down
Serial0/0/2              unassigned         down       down
Serial0/0/3              unassigned         down       down
[R2]
[R2]
<R2>ping 24.23.5.202
  PING 24.23.5.202: 56 data bytes, press CTRL C to break
    Reply from 24.23.5.202: bytes=56 Sequence=1 ttl=255 time=30 ms
    Reply from 24.23.5.202: bytes=56 Sequence=2 ttl=255 time=40 ms
    Reply from 24.23.5.202: bytes=56 Sequence=3 ttl=255 time=50 ms
    Reply from 24.23.5.202: bytes=56 Sequence=4 ttl=255 time=50 ms
    Reply from 24.23.5.202: bytes=56 Sequence=5 ttl=255 time=50 ms

  --- 24.23.5.202 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 30/44/50 ms
```

Router 3:

```
R3
[R3]interface G
[R3]interface GigabitEthernet 0/0/1
[R3-GigabitEthernet0/0/1]ip address 24.23.5.205 30
[R3-GigabitEthernet0/0/1]quit
[R3]display ip interface brief
*down: administratively down
!down: FIB overload down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
The number of interface that is UP in Physical is 2
The number of interface that is DOWN in Physical is 9
The number of interface that is UP in Protocol is 2
The number of interface that is DOWN in Protocol is 9

Interface                IP Address/Mask    Physical    Protocol
Ethernet0/0/0            unassigned         down       down
Ethernet0/0/1            unassigned         down       down
GigabitEthernet0/0/0     unassigned         down       down
GigabitEthernet0/0/1     24.23.5.205/30    up         up
GigabitEthernet0/0/2     unassigned         down       down
GigabitEthernet0/0/3     unassigned         down       down
NULL0                    unassigned         up         up(s)
Serial0/0/0              unassigned         down       down
Serial0/0/1              unassigned         down       down
Serial0/0/2              unassigned         down       down
Serial0/0/3              unassigned         down       down
[R3]save
^
Error: Unrecognized command found at '^' position.
[R3]quit
<R3>save
The current configuration will be written to the device.
Are you sure to continue?[Y/N]Y
Info: Please input the file name ( *.cfg, *.zip ) [vrpcfg.zip]:
Now saving the current configuration to the slot 17.
Save the configuration successfully.
<R3>
```

Router 4:

```
R4
[Huawei]interface GigabitEthernet 0/0/2
[Huawei-GigabitEthernet0/0/2]ip address 24.23.5.209 30
[Huawei-GigabitEthernet0/0/2]quit
[Huawei]display ip interface brief
*down: administratively down
!down: FIB overload down
^down: standby
(1): loopback
(s): spoofing
(d): Dampening Suppressed
The number of interface that is UP in Physical is 2
The number of interface that is DOWN in Physical is 9
The number of interface that is UP in Protocol is 2
The number of interface that is DOWN in Protocol is 9

Interface                IP Address/Mask      Physical  Protocol
Ethernet0/0/0            unassigned           down     down
Ethernet0/0/1            unassigned           down     down
GigabitEthernet0/0/0     unassigned           down     down
GigabitEthernet0/0/1     unassigned           down     down
GigabitEthernet0/0/2     24.23.5.209/30      up       up
GigabitEthernet0/0/3     unassigned           down     down
NULL0                   unassigned           up       up(s)
Serial0/0/0              unassigned           down     down
Serial0/0/1              unassigned           down     down
Serial0/0/2              unassigned           down     down
Serial0/0/3              unassigned           down     down
[Huawei]quit
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R4
[R4]quit
<R4>save
The current configuration will be written to the device.
Are you sure to continue?[Y/N]Y
Info: Please input the file name ( *.cfg, *.zip ) [vrpcfg.zip]:
Now saving the current configuration to the slot 17.
Save the configuration successfully.
<R4>
```

Router 5:

```
R5
(d): Dampening Suppressed
The number of interface that is UP in Physical is 2
The number of interface that is DOWN in Physical is 9
The number of interface that is UP in Protocol is 2
The number of interface that is DOWN in Protocol is 9

Interface                IP Address/Mask      Physical  Protocol
Ethernet0/0/0            unassigned           down     down
Ethernet0/0/1            unassigned           down     down
GigabitEthernet0/0/0     unassigned           down     down
GigabitEthernet0/0/1     unassigned           down     down
GigabitEthernet0/0/2     unassigned           down     down
GigabitEthernet0/0/3     24.23.5.213/30      up       up
NULL0                   unassigned           up       up(s)
Serial0/0/0              unassigned           down     down
Serial0/0/1              unassigned           down     down
Serial0/0/2              unassigned           down     down
Serial0/0/3              unassigned           down     down
[Huawei]sysname R5
[R5]quit
<R5>save
The current configuration will be written to the device.
Are you sure to continue?[Y/N]Y
Info: Please input the file name ( *.cfg, *.zip ) [vrpcfg.zip]:
Now saving the current configuration to the slot 17.
Save the configuration successfully.
<R5>ping 24.23.5.214
PING 24.23.5.214: 56 data bytes, press CTRL_C to break
  Reply from 24.23.5.214: bytes=56 Sequence=1 ttl=255 time=30 ms
  Reply from 24.23.5.214: bytes=56 Sequence=2 ttl=255 time=70 ms
  Reply from 24.23.5.214: bytes=56 Sequence=3 ttl=255 time=50 ms
  Reply from 24.23.5.214: bytes=56 Sequence=4 ttl=255 time=50 ms
  Reply from 24.23.5.214: bytes=56 Sequence=5 ttl=255 time=30 ms

--- 24.23.5.214 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 30/46/70 ms
<R5>
```

Lab-03 (CCN)

FRIDAY, JANUARY 30, 2026

NED University

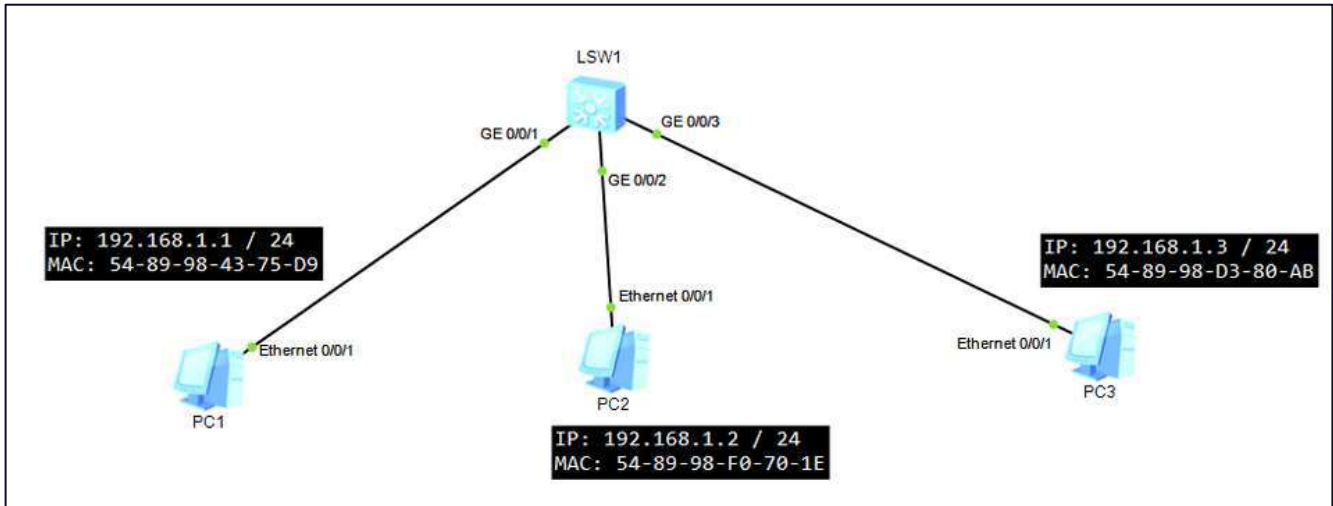
Name: MUHAMMAD OBAID

Roll Number : CT-23025 (Section A)

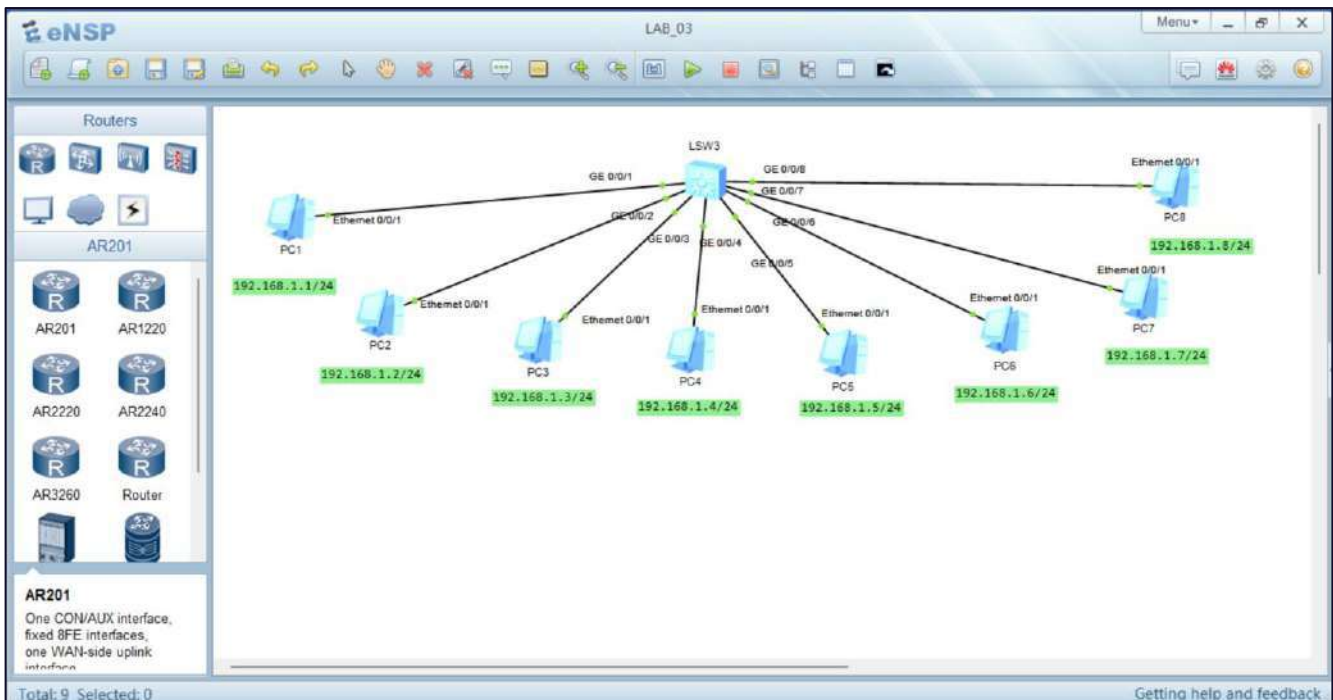


Lab Task (Lab # 03)

Create the following scenario in eNSP with the mentioned IP addresses. Please note the MAC address may vary system to system, mention your MAC addresses. Ping PC3 from PC1 and explain how the ARP table would be maintained and what would be the role of PC2 in this process. Also explain the difference between the static and dynamic ARP entries.



ENSP Simulator Work:



PC1

Basic Config Command MCPacket UdpPacket Console

Host Name:

MAC Address:

IPv4 Configuration

☒ Static ☐ DHCP ☐ Obtain DNS server address automatically

IP Address: DNS1:

Subnet Mask: DNS2:

Gateway:

PC3

Basic Config Command MCPacket UdpPacket Console

Welcome to use PC Simulator!

PC>ping 192.168.1.1

Ping 192.168.1.1: 32 data bytes, Press Ctrl_C to break

From 192.168.1.1: bytes=32 seq=1 ttl=128 time=62 ms

From 192.168.1.1: bytes=32 seq=2 ttl=128 time=47 ms

From 192.168.1.1: bytes=32 seq=3 ttl=128 time=47 ms

From 192.168.1.1: bytes=32 seq=4 ttl=128 time=47 ms

From 192.168.1.1: bytes=32 seq=5 ttl=128 time=47 ms

--- 192.168.1.1 ping statistics ---

5 packet(s) transmitted

5 packet(s) received

0.00% packet loss

round-trip min/avg/max = 47/50/62 ms

PC>arp -a

Internet Address	Physical Address	Type
192.168.1.1	54-89-98-00-27-BA	dynamic

PC>

PC1

Basic Config Command MCPacket UdpPacket Console

Welcome to use PC Simulator!

PC>arp -a

Internet Address	Physical Address	Type
192.168.1.3	54-89-98-76-62-FA	dynamic

PC>

PC5

Basic Config
Command
MCPacket
UdpPacket
Console

Host Name:
pc5

MAC Address:
54-89-98-F4-04-BD

IPv4 Configuration

☒ Static
☐ DHCP
☐ Obtain DNS server address automatically

IP Address:
192 . 168 . 1 . 5
DNS1:
0 . 0 . 0 . 0

Subnet Mask:
255 . 255 . 255 . 0
DNS2:
0 . 0 . 0 . 0

Gateway:
0 . 0 . 0 . 0

IPv6 Configuration

☒ Static
☐ DHCPv6

IPv6 Address:
::

Prefix Length:
128

IPv6 Gateway:
::

Apply

PC7

Basic Config
Command
MCPacket
UdpPacket
Console

```

Ping 192.168.1.5: 32 data bytes, Press Ctrl_C to break
From 192.168.1.5: bytes=32 seq=1 ttl=128 time=47 ms
From 192.168.1.5: bytes=32 seq=2 ttl=128 time=47 ms
From 192.168.1.5: bytes=32 seq=3 ttl=128 time=63 ms
From 192.168.1.5: bytes=32 seq=4 ttl=128 time=47 ms
From 192.168.1.5: bytes=32 seq=5 ttl=128 time=47 ms

--- 192.168.1.5 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 47/50/63 ms

PC>ping 192.168.1.1

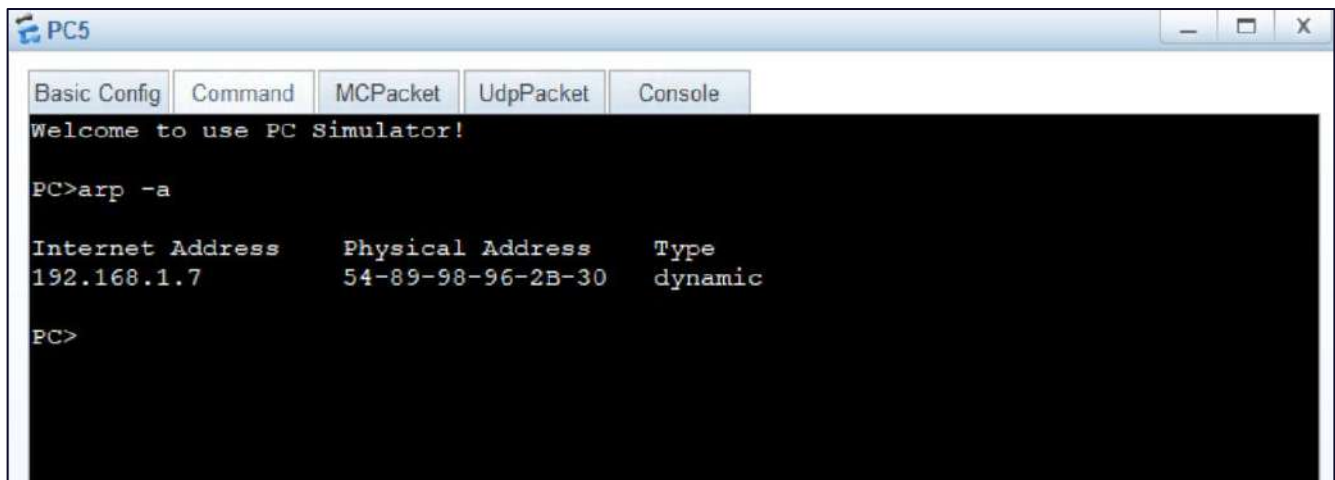
Ping 192.168.1.1: 32 data bytes, Press Ctrl_C to break
From 192.168.1.1: bytes=32 seq=1 ttl=128 time=47 ms
From 192.168.1.1: bytes=32 seq=2 ttl=128 time=47 ms
From 192.168.1.1: bytes=32 seq=3 ttl=128 time=47 ms
From 192.168.1.1: bytes=32 seq=4 ttl=128 time=47 ms
From 192.168.1.1: bytes=32 seq=5 ttl=128 time=47 ms

--- 192.168.1.1 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 47/47/47 ms

PC>arp -a

Internet Address      Physical Address      Type
192.168.1.1           54-89-98-00-27-BA    dynamic
192.168.1.5           54-89-98-F4-04-BD    dynamic

```



The screenshot shows a window titled 'PC5' with a menu bar containing 'Basic Config', 'Command', 'MCPacket', 'UdpPacket', and 'Console'. The 'Console' tab is active, displaying the following text:

```
Welcome to use PC Simulator!

PC>arp -a

Internet Address      Physical Address      Type
192.168.1.7           54-89-98-96-2B-30     dynamic

PC>
```

ARP (Address Resolution Protocol) maps an **IP address** → **MAC address** and stores the result in an ARP table.

Static ARP Entry is manually configured by an administrator

Key traits:

- Added manually (e.g. `arp -s`)
- **Does not expire**
- Survives until removed or the system reboots (depends on OS)
- Not learned from ARP traffic

While Dynamic ARP Entry is automatically learned via ARP requests/replies

Key traits:

- Created automatically when a device communicates
- **Has a timeout** (ages out if unused)
- Updates itself if MAC addresses change
- Default behavior on most networks

Lab-04 (CCN)

THURSDAY, FEBRUARY 12, 2026

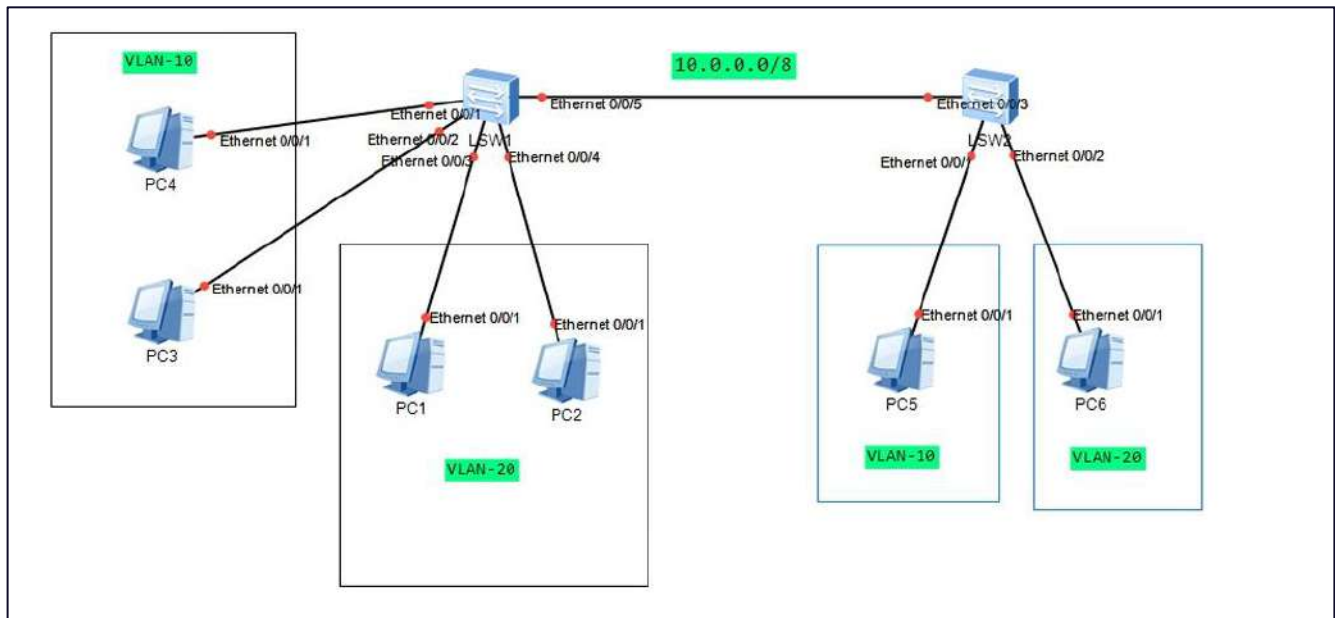
NED University

Name: MUHAMMAD OBAID

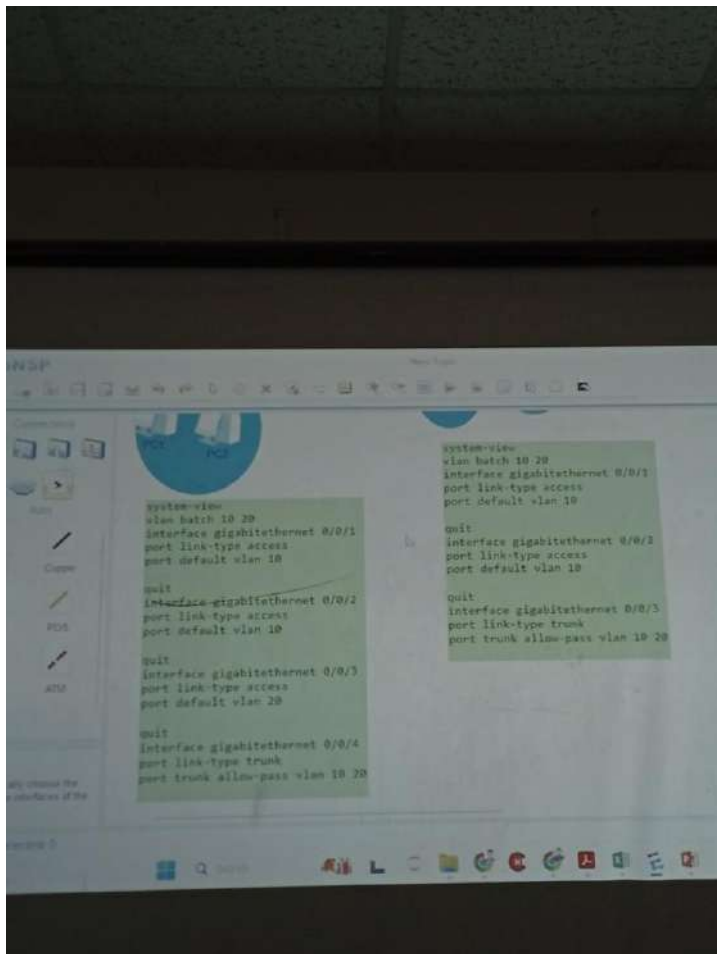
Roll Number : CT-23025 (Section A)

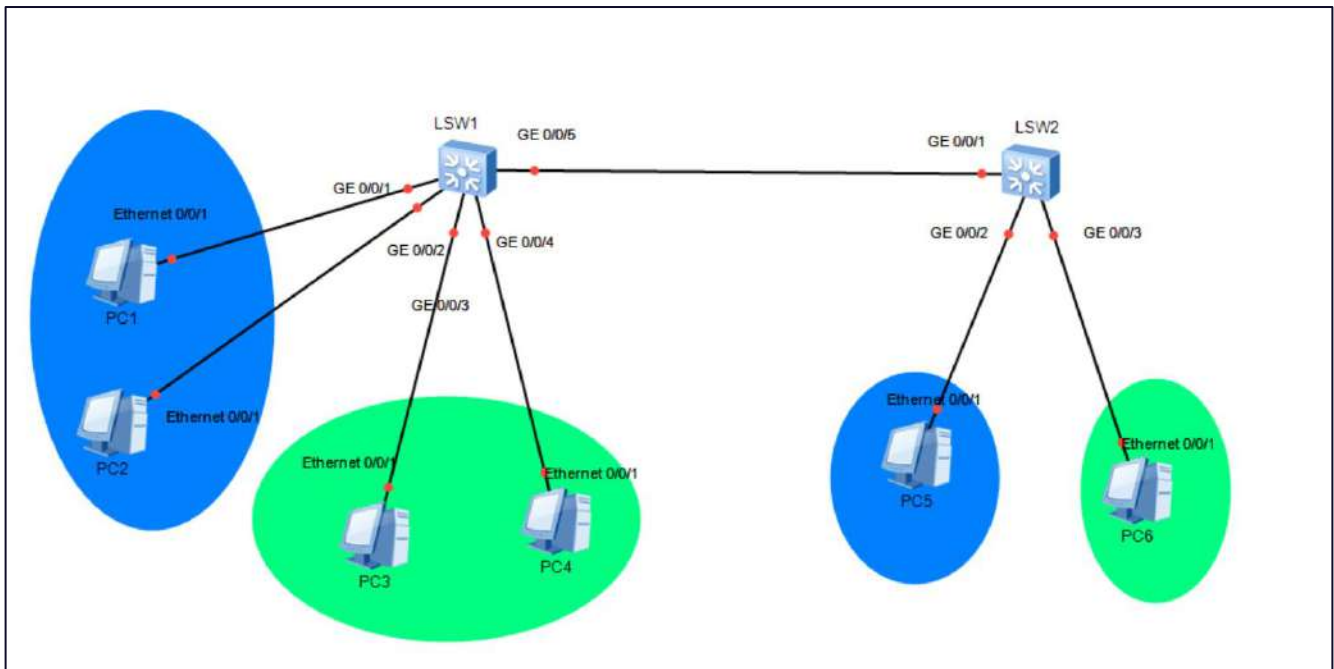


Implement the following VLAN scenario in eNSP simulator:



ESPN Diagram:





Making request from Pc3 → Pc6:

```

PC3
Basic Config Command MCPacket UdpPacket Console
Welcome to use PC Simulator!

PC>ping 192.168.20.3

Ping 192.168.20.3: 32 data bytes, Press Ctrl_C to break
From 192.168.20.1: Destination host unreachable
From 192.168.20.1: Destination host unreachable
From 192.168.20.1: Destination host unreachable
From 192.168.20.3: bytes=32 seq=1 ttl=128 time=62 ms
From 192.168.20.3: bytes=32 seq=2 ttl=128 time=62 ms
From 192.168.20.3: bytes=32 seq=3 ttl=128 time=47 ms
From 192.168.20.3: bytes=32 seq=4 ttl=128 time=94 ms
From 192.168.20.3: bytes=32 seq=5 ttl=128 time=78 ms

--- 192.168.20.3 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 47/68/94 ms

PC>arp -a

Internet Address      Physical Address      Type
192.168.20.3          54-89-98-66-0A-1B    dynamic
PC>

PC6
Basic Config Command MCPacket UdpPacket Console
Welcome to use PC Simulator!

PC>arp -a

Internet Address      Physical Address      Type
192.168.20.1          54-89-98-DE-10-69    dynamic

```

Lab-05 (CCN)

THURSDAY, FEBRUARY 19, 2026

NED University

Name: MUHAMMAD OBAID

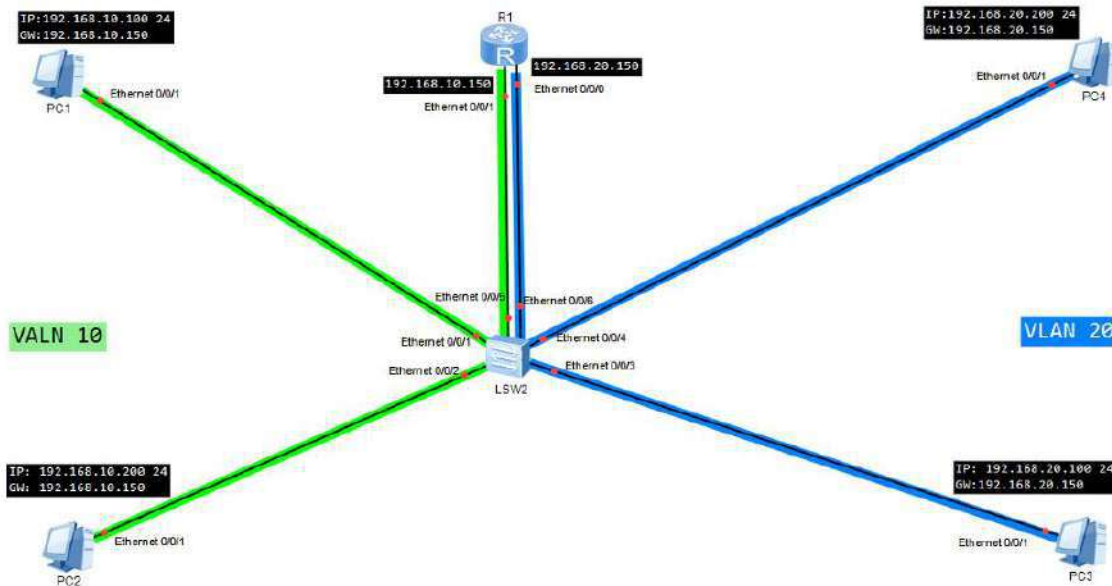
Roll Number : CT-23025 (Section A)



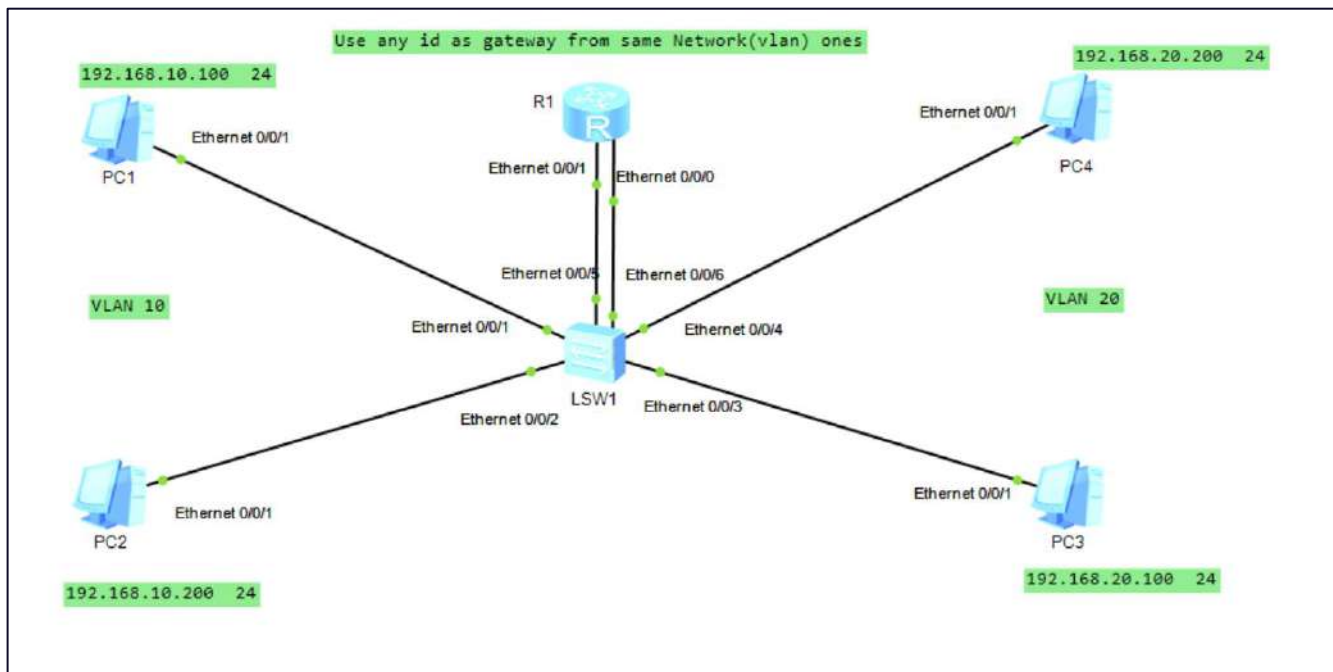
Implement the following VLAN scenario in eNSP simulator:

Lab Task (Lab # 05):

Configure the following scenario in eNSP:



Configuration:



For Switch:

LSW1

The device is running!

```
<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]interface Eth
[Huawei]interface Ethernet 0/0/1
[Huawei-Ethernet0/0/1]quit
[Huawei]vlan batch 10 20
Info: This operation may take a few seconds. Please wait for a moment...done.
[Huawei]interface Ethernet 0/0/1
[Huawei-Ethernet0/0/1]port link-type access
[Huawei-Ethernet0/0/1]port default vlan 10
[Huawei-Ethernet0/0/1]quit
[Huawei]interface Ethernet 0/0/2
[Huawei-Ethernet0/0/2]port link-type access
[Huawei-Ethernet0/0/2]port default vlan 10
[Huawei-Ethernet0/0/2]quit
[Huawei]interface Ethernet 0/0/5
[Huawei-Ethernet0/0/5]port link-type access
[Huawei-Ethernet0/0/5]port default vlan 10
[Huawei-Ethernet0/0/5]quit
[Huawei]interface Ethernet 0/0/3
[Huawei-Ethernet0/0/3]port link-type access
[Huawei-Ethernet0/0/3]port default vlan 20
[Huawei-Ethernet0/0/3]quit
[Huawei]interface Ethernet 0/0/4
[Huawei-Ethernet0/0/4]port link-type access
[Huawei-Ethernet0/0/4]port default vlan 20
[Huawei-Ethernet0/0/4]quit
[Huawei]interface Ethernet 0/0/6
[Huawei-Ethernet0/0/6]port link-type access
[Huawei-Ethernet0/0/6]port default vlan 20
[Huawei-Ethernet0/0/6]quit
[Huawei]display interface ip brief
      ^
Error: Wrong parameter found at '^' position.
[Huawei]display ip interface brief
*down: administratively down
*down: standby
(l): loopback
(s): spoofing
The number of interface that is UP in Physical is 1
The number of interface that is DOWN in Physical is 2
The number of interface that is UP in Protocol is 1
The number of interface that is DOWN in Protocol is 2

Interface                IP Address/Mask      Physical  Protocol
Meth0/0/1                unassigned           down      down
NULL0                    unassigned           up        up(s)
Vlanif1                  unassigned           down      down
[Huawei]quit
<Huawei>save
The current configuration will be written to the device.
Are you sure to continue?[Y/N]Y
Info: Please input the file name ( *.cfg, *.zip ) [vrpcfg.zip]:
Now saving the current configuration to the slot 0.
Save the configuration successfully.
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname LSW1
[LSW1]quit
<LSW1>save
The current configuration will be written to the device.
Are you sure to continue?[Y/N]Y
Now saving the current configuration to the slot 0.
Save the configuration successfully.
<LSW1>
```

For Router:

```

R1
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]
Feb 19 2026 16:11:36-08:00 R1 DS/4/DATASYNC_CFGCHANGE:OID 1.3.6.1.4.1.2011.5.25.
191.3.1 configurations have been changed. The current change number is 1, the ch
ange loop count is 0, and the maximum number of records is 4095.
[R1]quit
<R1>u t m
Info: Current terminal monitor is off.
<R1>system-view
Enter system view, return user view with Ctrl+Z.
[R1]interface Eth
[R1]interface Ethernet 0/0/1
[R1-Ethernet0/0/1]ip address 192.168.10.150 24
[R1-Ethernet0/0/1]quit
[R1]interface Ethernet 0/0/0
[R1-Ethernet0/0/0]ip address 192.168.20.150 24
[R1-Ethernet0/0/0]quit
[R1]display interface ip brief
      ^
Error: Wrong parameter found at '^' position.
[R1]display ip interface brief
*down: administratively down
!down: FIB overload down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
The number of interface that is UP in Physical is 3
The number of interface that is DOWN in Physical is 8
The number of interface that is UP in Protocol is 3
The number of interface that is DOWN in Protocol is 8

Interface                IP Address/Mask      Physical      Protocol
Ethernet0/0/0            192.168.20.150/24    up           up
Ethernet0/0/1            192.168.10.150/24    up           up
GigabitEthernet0/0/0     unassigned           down         down
GigabitEthernet0/0/1     unassigned           down         down
GigabitEthernet0/0/2     unassigned           down         down

```

Ping Output:

```

PC2
Basic Config  Command  MCPacket  UdpPacket  Console
Welcome to use PC Simulator!

PC>ping 192.168.10.100

Ping 192.168.10.100: 32 data bytes, Press Ctrl_C to break
From 192.168.10.100: bytes=32 seq=1 ttl=128 time=31 ms
From 192.168.10.100: bytes=32 seq=2 ttl=128 time=47 ms
From 192.168.10.100: bytes=32 seq=3 ttl=128 time=47 ms
From 192.168.10.100: bytes=32 seq=4 ttl=128 time=47 ms
From 192.168.10.100: bytes=32 seq=5 ttl=128 time=47 ms

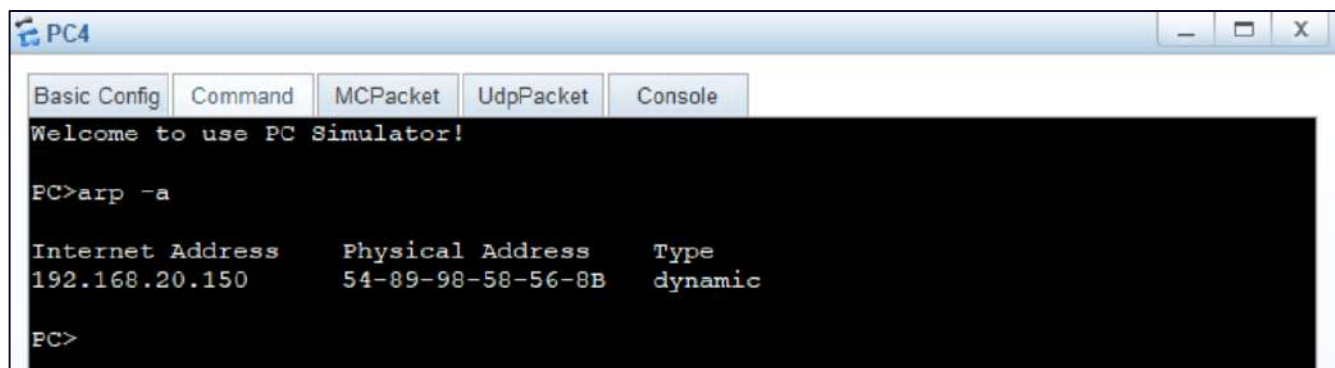
--- 192.168.10.100 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 31/43/47 ms

PC>ping 192.168.20.200

Ping 192.168.20.200: 32 data bytes, Press Ctrl_C to break
From 192.168.20.200: bytes=32 seq=1 ttl=127 time=156 ms
From 192.168.20.200: bytes=32 seq=2 ttl=127 time=94 ms
From 192.168.20.200: bytes=32 seq=3 ttl=127 time=94 ms
From 192.168.20.200: bytes=32 seq=4 ttl=127 time=125 ms
From 192.168.20.200: bytes=32 seq=5 ttl=127 time=94 ms

--- 192.168.20.200 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 94/112/156 ms

```



Lab-06 (CCN)

THURSDAY, FEBRUARY 19, 2026

NED University

Name: MUHAMMAD OBAID

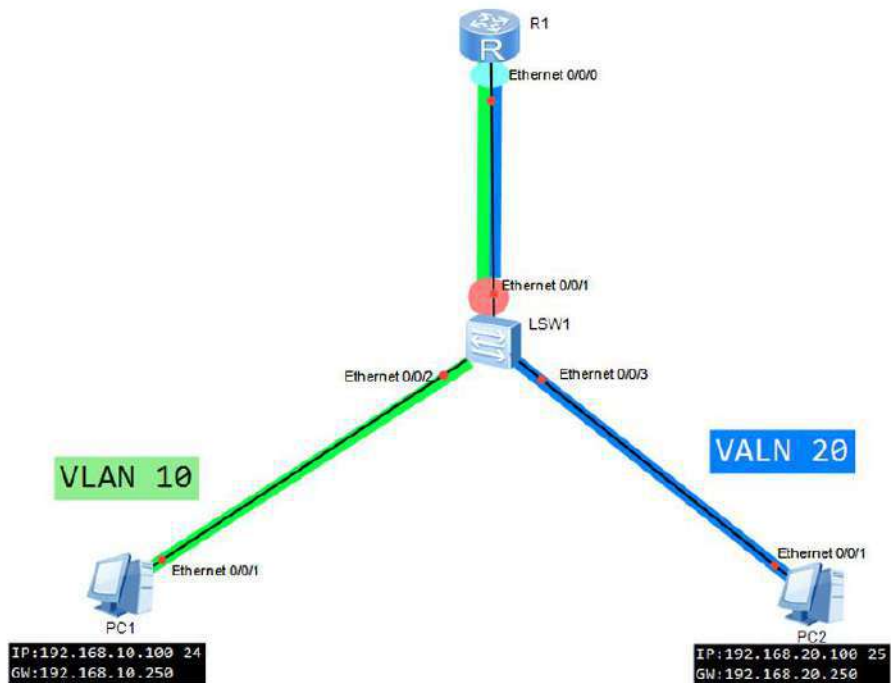
Roll Number : CT-23025 (Section A)



Configure the following router-on-a-stick in eNSP:

Lab Task (Lab # 06):

Configure the following router-on-a-stick in eNSP.

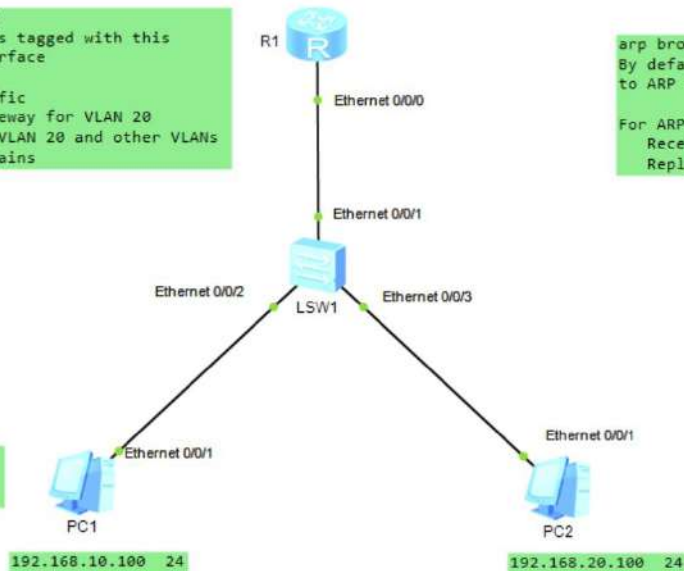


Configuration:

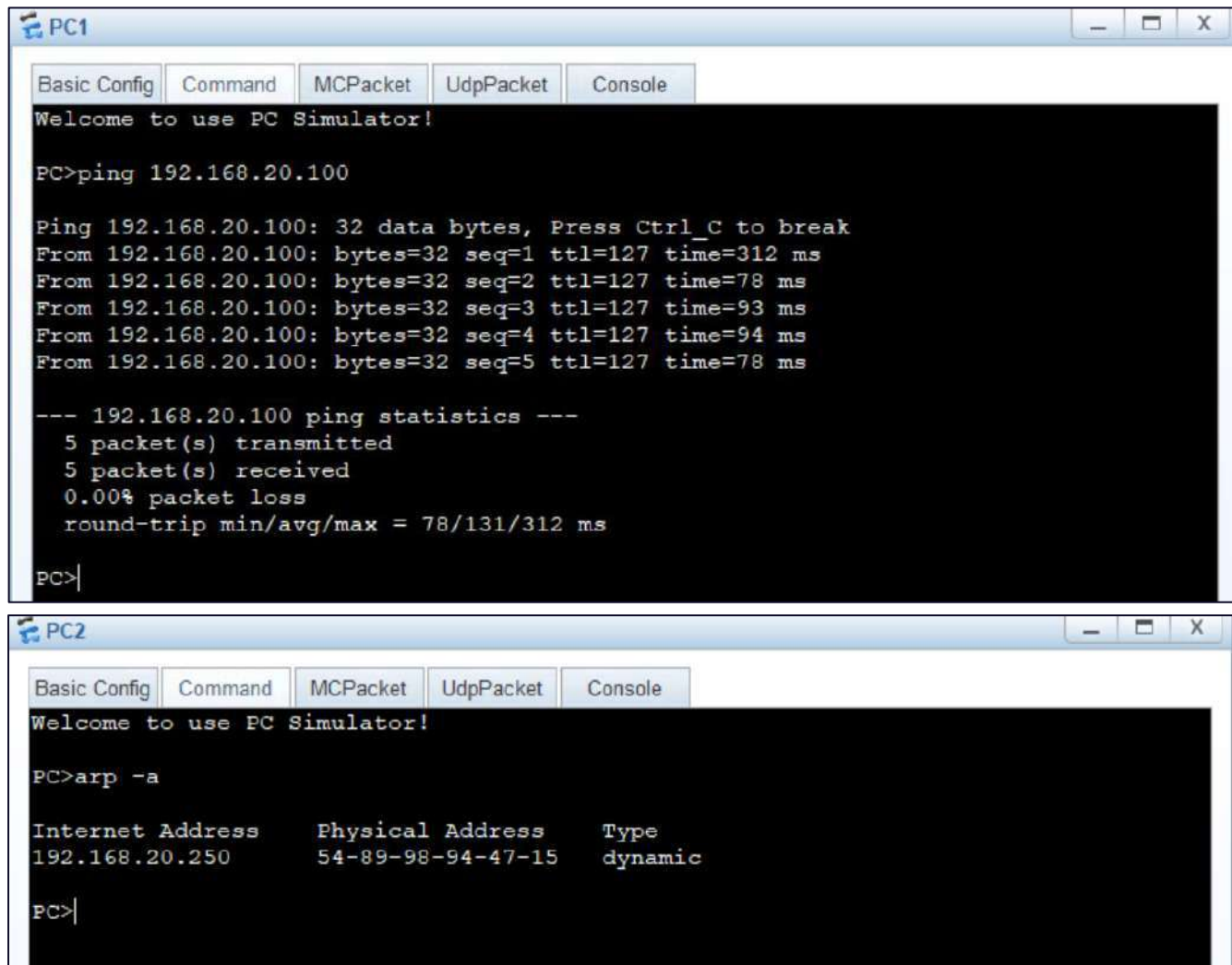
```
dot1q termination vid 20
Accept and process frames tagged with this
VLAN ID on this sub-interface
It is used to:
  Terminate VLAN 20 traffic
  Act as the default gateway for VLAN 20
  Route traffic between VLAN 20 and other VLANs
  Separate broadcast domains
```

```
arp broadcast enable
By default, the router will not respond
to ARP requests on that sub-interface
For ARP-broadcast router must:
  Receive it
  Reply with its MAC address
```

```
Must give gateway-IP
at the end for
not-reachable mistake
```



Ping Output:



The image shows two windows from a PC simulator. The top window, titled 'PC1', has tabs for 'Basic Config', 'Command', 'MCPacket', 'UdpPacket', and 'Console'. The 'Console' tab is active, displaying the following text:

```
Welcome to use PC Simulator!

PC>ping 192.168.20.100

Ping 192.168.20.100: 32 data bytes, Press Ctrl_C to break
From 192.168.20.100: bytes=32 seq=1 ttl=127 time=312 ms
From 192.168.20.100: bytes=32 seq=2 ttl=127 time=78 ms
From 192.168.20.100: bytes=32 seq=3 ttl=127 time=93 ms
From 192.168.20.100: bytes=32 seq=4 ttl=127 time=94 ms
From 192.168.20.100: bytes=32 seq=5 ttl=127 time=78 ms

--- 192.168.20.100 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 78/131/312 ms

PC>
```

The bottom window, titled 'PC2', also has the same tabs. The 'Console' tab is active, displaying the following text:

```
Welcome to use PC Simulator!

PC>arp -a

Internet Address      Physical Address      Type
192.168.20.250        54-89-98-94-47-15    dynamic

PC>
```

Lab-07 (CCN)

THURSDAY, FEBRUARY 19, 2026

NED University

Name: MUHAMMAD OBAID

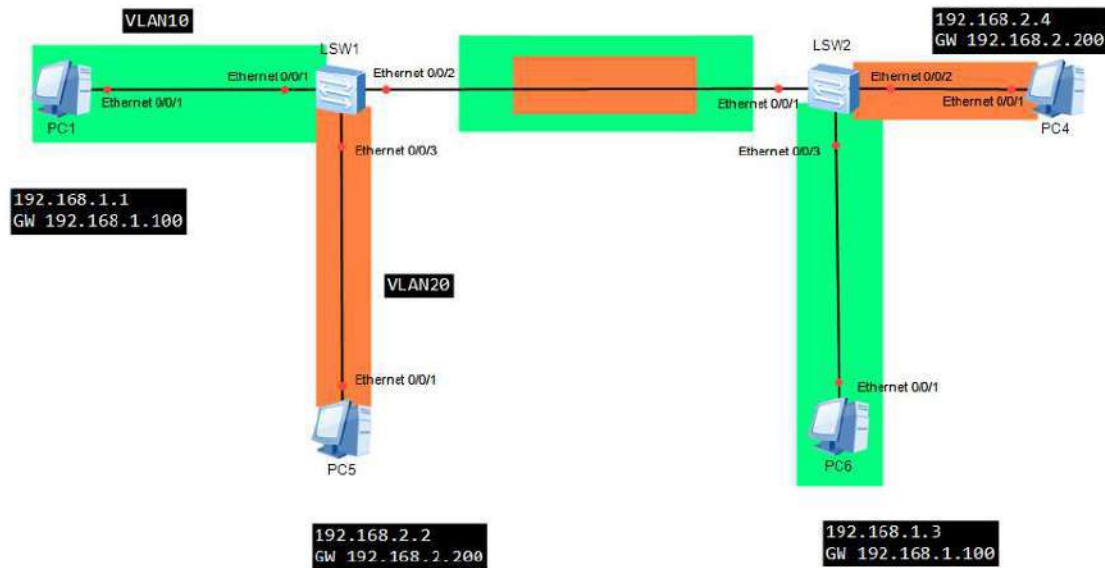
Roll Number : CT-23025 (Section A)



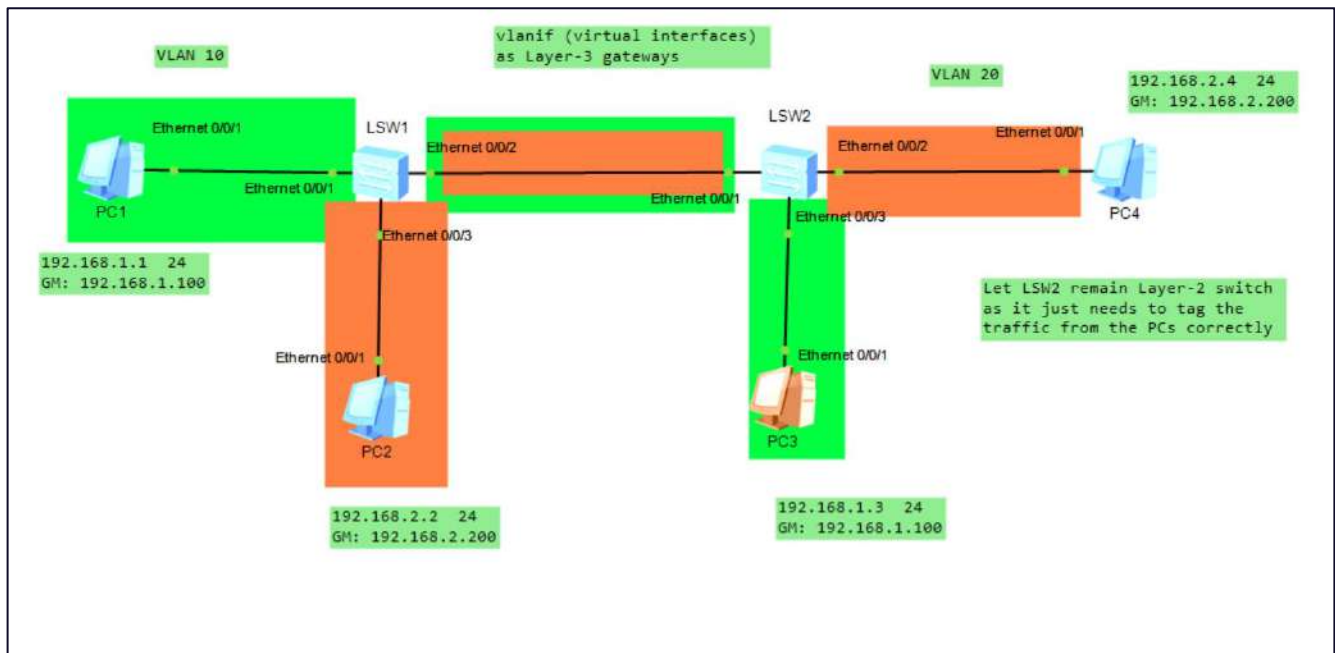
Configure the following scenario (Inter VLAN routing) in eNSP:

Lab Task (Lab # 07):

Configure the following scenario (Inter VLAN routing) in eNSP.



Configuration:



For Switch (LSW1):-

```

LSW1
The device is running!

<Huawei>
<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname LSW1
[LSW1]vlan batch 10 20
Info: This operation may take a few seconds. Please wait for a moment...done.
[LSW1]interface Eth
[LSW1]interface Ethernet 0/0/1
[LSW1-Ethernet0/0/1]port link-type access
[LSW1-Ethernet0/0/1]port default vlan 10
[LSW1-Ethernet0/0/1]quit
[LSW1]interface Ethernet 0/0/3
[LSW1-Ethernet0/0/3]port link-type access
[LSW1-Ethernet0/0/3]port default vlan 20
[LSW1-Ethernet0/0/3]quit
[LSW1]interface Ethernet 0/0/2
[LSW1-Ethernet0/0/2]port link-type trunk
[LSW1-Ethernet0/0/2]port trunk allow-pass vlan 10 20
[LSW1-Ethernet0/0/2]quit
[LSW1]display ip interface brief
*down: administratively down
^down: standby
(l): loopback
(s): spoofing
The number of interface that is UP in Physical is 2
The number of interface that is DOWN in Physical is 1
The number of interface that is UP in Protocol is 1
The number of interface that is DOWN in Protocol is 2

Interface                IP Address/Mask      Physical  Protocol
MEth0/0/1                unassigned           down     down
NULL0                   unassigned           up       up(s)
Vlanif1                 unassigned           up       down
[LSW1]interface Vlanif 10
[LSW1-Vlanif10]ip address 192.168.1.100
^
Error:Incomplete command found at '^' position.
[LSW1-Vlanif10]ip address 192.168.1.100 24
[LSW1-Vlanif10]quit
[LSW1]interface Vlanif 20
[LSW1-Vlanif20]ip address 192.168.2.200 24
[LSW1-Vlanif20]quit
[LSW1]quit
<LSW1>save
The current configuration will be written to the device.
Are you sure to continue?[Y/N]Y
Info: Please input the file name ( *.cfg, *.zip ) [vrpcfg.zip]:
Now saving the current configuration to the slot 0.
Save the configuration successfully.
<LSW1>

```

For Switch (LSW2):

```

LSW2
The device is running!

<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname LSW2
[LSW2]vlan batch 10 20
Info: This operation may take a few seconds. Please wait for a moment...done.
[LSW2]interface Eth
[LSW2]interface Ethernet 0/0/3
[LSW2-Ethernet0/0/3]port link-type access
[LSW2-Ethernet0/0/3]port default vlan 10
[LSW2-Ethernet0/0/3]quit
[LSW2]interface Ethernet 0/0/2
[LSW2-Ethernet0/0/2]port link-type access
[LSW2-Ethernet0/0/2]port default vlan 20
[LSW2-Ethernet0/0/2]quit
[LSW2]interface Ethernet 0/0/1
[LSW2-Ethernet0/0/1]port link-type trunk
[LSW2-Ethernet0/0/1]port trunk allow-pass vlan 10 20
[LSW2-Ethernet0/0/1]quit
[LSW2]quit
<LSW2>save
The current configuration will be written to the device.
Are you sure to continue?[Y/N]Y
Info: Please input the file name ( *.cfg, *.zip ) [vrpcfg.zip]:
Now saving the current configuration to the slot 0.
Save the configuration successfully.
<LSW2>

```

Ping Output:

```

PC1
Basic Config Command MCPacket UdpPacket Console
Ping 192.168.2.2: 32 data bytes, Press Ctrl_C to break
From 192.168.2.2: bytes=32 seq=1 ttl=127 time=78 ms
From 192.168.2.2: bytes=32 seq=2 ttl=127 time=63 ms
From 192.168.2.2: bytes=32 seq=3 ttl=127 time=62 ms
From 192.168.2.2: bytes=32 seq=4 ttl=127 time=63 ms
From 192.168.2.2: bytes=32 seq=5 ttl=127 time=47 ms

--- 192.168.2.2 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 47/62/78 ms

PC>ping 192.168.1.3

Ping 192.168.1.3: 32 data bytes, Press Ctrl_C to break
From 192.168.1.3: bytes=32 seq=1 ttl=128 time=78 ms
From 192.168.1.3: bytes=32 seq=2 ttl=128 time=47 ms
From 192.168.1.3: bytes=32 seq=3 ttl=128 time=110 ms
From 192.168.1.3: bytes=32 seq=4 ttl=128 time=78 ms
From 192.168.1.3: bytes=32 seq=5 ttl=128 time=62 ms

--- 192.168.1.3 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 47/75/110 ms

PC>arp -a

Internet Address    Physical Address    Type
192.168.1.100       4C-1F-CC-10-48-13  dynamic
192.168.1.3         54-89-98-92-1C-0F  dynamic
PC>

```

```

PC3
Basic Config Command MCPacket UdpPacket Console
Welcome to use PC Simulator!

PC>arp -a

Internet Address    Physical Address    Type
192.168.1.1         54-89-98-B3-23-29  dynamic
PC>

```

Lab 8 (CCN)

TUESDAY, APRIL 7, 2026

NED University

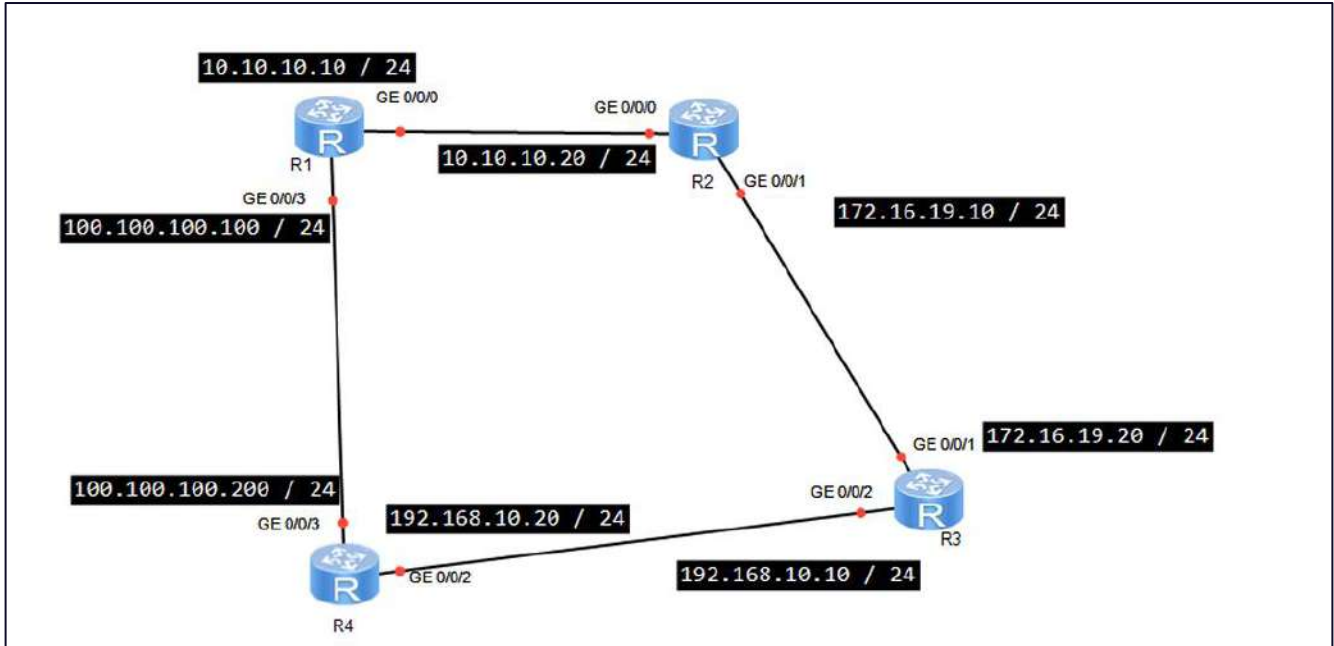
Name: MUHAMMAD OBAID

Roll Number : CT-23025 (Section A)

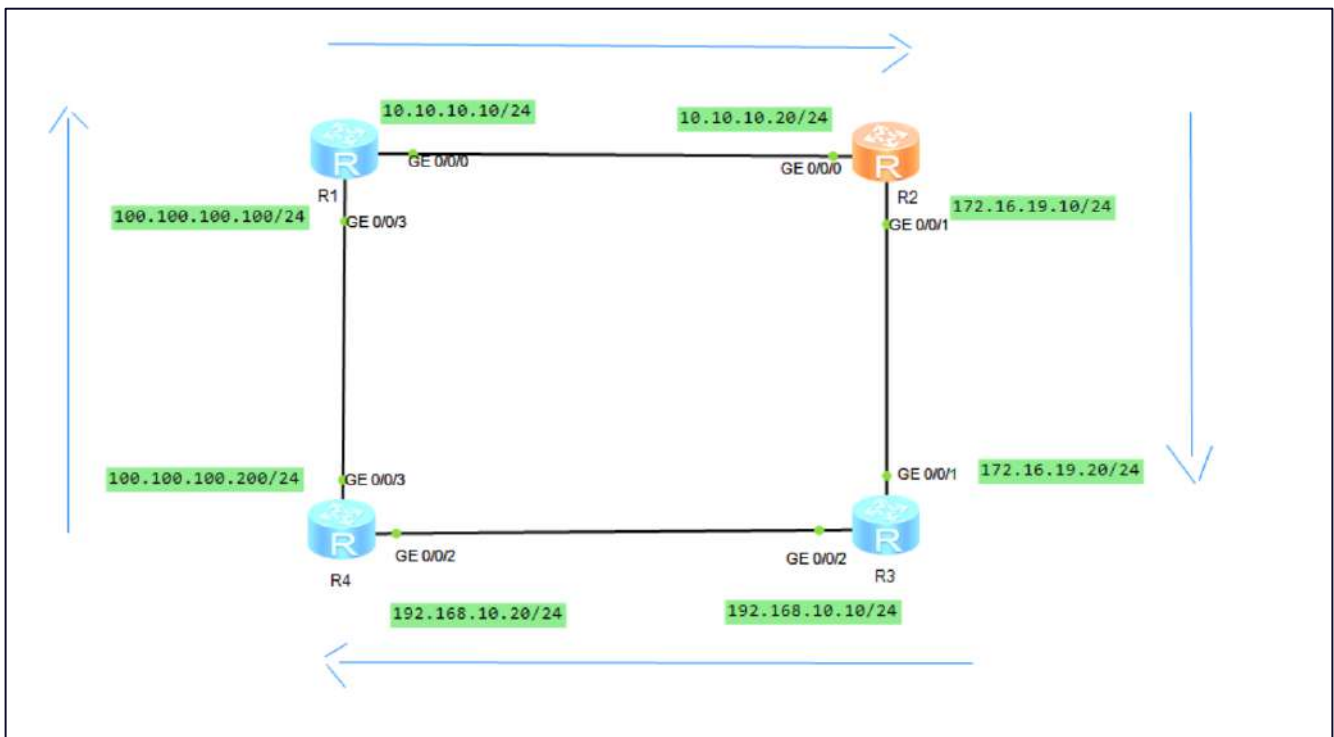


Lab Task (Lab # 08) :

Configure the following scenario by applying static routes in eNSP. Note: All routers should be able to ping all interfaces



Configuration with Ping:



```
R1
<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>sys
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]inter
[Huawei]interface G
[Huawei]interface GigabitEthernet 0/0/0
[Huawei-GigabitEthernet0/0/0]ip address 10.10.10.10 24
[Huawei-GigabitEthernet0/0/0]quit
[Huawei]interface GigabitEthernet 0/0/3
[Huawei-GigabitEthernet0/0/3]ip address 100.100.100.100 24
[Huawei-GigabitEthernet0/0/3]quit
[Huawei]sysname R1
[R1]ip route-static 172.16.19.0 24 10.10.10.20
[R1]ip route-static 192.168.10.20 24 10.10.10.20
Info: The destination address and mask of the configured static route mismatched
, and the static route 192.168.10.0/24 was generated.
[R1]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
Destinations : 8          Routes : 8

Destination/Mask    Proto    Pre  Cost    Flags NextHop         Interface
-----
10.10.10.0/24      Direct   0     0        D  10.10.10.10      GigabitEthernet
0/0/0
10.10.10.10/32     Direct   0     0        D  127.0.0.1        GigabitEthernet
0/0/0
100.100.100.0/24   Direct   0     0        D  100.100.100.100  GigabitEthernet
0/0/3
100.100.100.100/32 Direct   0     0        D  127.0.0.1        GigabitEthernet
0/0/3
127.0.0.0/8        Direct   0     0        D  127.0.0.1        InLoopBack0
127.0.0.1/32       Direct   0     0        D  127.0.0.1        InLoopBack0
172.16.19.0/24     Static   60    0        RD  10.10.10.20      GigabitEthernet
0/0/0
192.168.10.0/24    Static   60    0        RD  10.10.10.20      GigabitEthernet
0/0/0

[R1]undo ip route-static 192.168.10.20 24 10.10.10.20
Warning: The destination address and the mask do not match, and thus the static
route 192.168.10.0/24 will be deleted. Continue?[Y/N]:N
[R1]quit
<R1>save
The current configuration will be written to the device.
Are you sure to continue?[Y/N]Y
Info: Please input the file name ( *.cfg, *.zip ) [vrpcfg.zip]:
Now saving the current configuration to the slot 17.
Save the configuration successfully.
<R1>tracert 192.168.10.20

Traceroute to 192.168.10.20(192.168.10.20), max hops: 30 ,packet length: 40,pr
ess CTRL_C to break

 1 10.10.10.20 80 ms  50 ms  50 ms
 2 172.16.19.20 120 ms  60 ms  80 ms
 3 192.168.10.20 90 ms  80 ms  100 ms
<R1>
```

```
R2
<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>sys
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]interface G
[Huawei]interface GigabitEthernet 0/0/0
[Huawei-GigabitEthernet0/0/0]ip address 10.10.10.20 24
[Huawei-GigabitEthernet0/0/0]quit
[Huawei]interface GigabitEthernet 0/0/1
[Huawei-GigabitEthernet0/0/1]ip address 172.16.19.10 24
[Huawei-GigabitEthernet0/0/1]quit
[Huawei]sysname R2
[R2]ip route-static 192.168.10.0 24 172.16.19.20
[R2]ip route-static 100.100.100.0 24 172.16.19.20
[R2]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
Destinations : 8          Routes : 8

Destination/Mask    Proto  Pre  Cost    Flags NextHop          Interface
-----
10.10.10.0/24      Direct  0    0        D  10.10.10.20          GigabitEthernet
0/0/0
10.10.10.20/32      Direct  0    0        D  127.0.0.1            GigabitEthernet
0/0/0
100.100.100.0/24    Static  60    0        RD  172.16.19.20          GigabitEthernet
0/0/1
127.0.0.0/8         Direct  0    0        D  127.0.0.1            InLoopBack0
127.0.0.1/32        Direct  0    0        D  127.0.0.1            InLoopBack0
172.16.19.0/24      Direct  0    0        D  172.16.19.10         GigabitEthernet
0/0/1
172.16.19.10/32     Direct  0    0        D  127.0.0.1            GigabitEthernet
0/0/1
192.168.10.0/24     Static  60    0        RD  172.16.19.20          GigabitEthernet
0/0/1

[R2]quit
<R2>save
The current configuration will be written to the device.
Are you sure to continue?[Y/N]Y
Info: Please input the file name ( *.cfg, *.zip ) [vrpcfg.zip]:
Now saving the current configuration to the slot 17.
Save the configuration successfully.
<R2>tracert 192.168.10.10

  traceroute to 192.168.10.10(192.168.10.10), max hops: 30 ,packet length: 40,pr
  ess CTRL_C to break

   1 172.16.19.20 30 ms  50 ms  50 ms
<R2>
```

```
R3
<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>sys
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]interface G
[Huawei]interface GigabitEthernet 0/0/1
[Huawei-GigabitEthernet0/0/1]ip address 172.16.19.20 24
[Huawei-GigabitEthernet0/0/1]quit
[Huawei]interface GigabitEthernet 0/0/2
[Huawei-GigabitEthernet0/0/2]ip address 192.168.10.10 24
[Huawei-GigabitEthernet0/0/2]quit
[Huawei]sysname R3
[R3]ip route-static 100.100.100.0 24 192.168.10.20
[R3]ip route-static 10.10.10.0 24 192.168.10.20
[R3]display ip routing-table protocol static
Route Flags: R - relay, D - download to fib
-----
Public routing table : Static
Destinations : 2          Routes : 2          Configured Routes : 2

Static routing table status : <Active>
Destinations : 2          Routes : 2

Destination/Mask    Proto  Pre  Cost    Flags NextHop          Interface
-----
10.10.10.0/24      Static  60    0        RD  192.168.10.20          GigabitEthernet
0/0/2
100.100.100.0/24    Static  60    0        RD  192.168.10.20          GigabitEthernet
0/0/2

Static routing table status : <Inactive>
Destinations : 0          Routes : 0

[R3]quit
<R3>save
The current configuration will be written to the device.
Are you sure to continue?[Y/N]Y
Info: Please input the file name ( *.cfg, *.zip ) [vrpcfg.zip]:
Now saving the current configuration to the slot 17.
```

```

Save the configuration successfully.
<R3>tracert 10.10.10.20

  traceroute to 10.10.10.20 (10.10.10.20), max hops: 30 ,packet length: 40,press
  CTRL_C to break

  1 192.168.10.20 50 ms  40 ms  50 ms

  2 100.100.100.100 60 ms  60 ms  50 ms

  3 10.10.10.20 50 ms  40 ms  70 ms
<R3>

```

```

R4
The device is running!

<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>sys
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]interface G
[Huawei]interface GigabitEthernet 0/0/2
[Huawei-GigabitEthernet0/0/2]ip address 192.168.10.20 24
[Huawei-GigabitEthernet0/0/2]quit
[Huawei]interface GigabitEthernet 0/0/3
[Huawei-GigabitEthernet0/0/3]ip address 100.100.100.200 24
[Huawei-GigabitEthernet0/0/3]quit
[Huawei]sysname R4
[R4]ping 100.100.100.100
  PING 100.100.100.100: 56  data bytes, press CTRL_C to break
    Reply from 100.100.100.100: bytes=56 Sequence=1 ttl=255 time=70 ms
    Reply from 100.100.100.100: bytes=56 Sequence=2 ttl=255 time=50 ms
    Reply from 100.100.100.100: bytes=56 Sequence=3 ttl=255 time=20 ms
    Reply from 100.100.100.100: bytes=56 Sequence=4 ttl=255 time=20 ms
    Reply from 100.100.100.100: bytes=56 Sequence=5 ttl=255 time=30 ms

  --- 100.100.100.100 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 20/38/70 ms

[R4]ip route-static 10.10.10.0 24 100.100.100.100
[R4]ip route-static 172.16.19.0 24 100.100.100.100
[R4]display ip routing-table protocol static
      ^
Error: Unrecognized command found at '^' position.
[R4]display ip routing-table protocol static
Route Flags: R - relay, D - download to fib
-----
Public routing table : Static
      Destinations : 2          Routes : 2          Configured Routes : 2

Destination/Mask    Proto  Pre  Cost    Flags NextHop         Interface
-----
10.10.10.0/24      Static 60   0       RD    100.100.100.100 GigabitEthernet
0/0/3
172.16.19.0/24     Static 60   0       RD    100.100.100.100 GigabitEthernet
0/0/3

Static routing table status : <Inactive>
      Destinations : 0          Routes : 0

[R4]quit
<R4>save
The current configuration will be written to the device.
Are you sure to continue?[Y/N]Y
Info: Please input the file name ( *.cfg, *.zip ) [vrpcfg.zip]:
Now saving the current configuration to the slot 17.
Save the configuration successfully.
<R4>tracert 172.16.19.10

  traceroute to 172.16.19.10 (172.16.19.10), max hops: 30 ,packet length: 40,press
  CTRL_C to break

  1 100.100.100.100 30 ms  40 ms  50 ms

  2 10.10.10.20 80 ms  50 ms  40 ms
<R4>

```

Static Routing (CCN)

SATURDAY, FEBRUARY 21, 2026

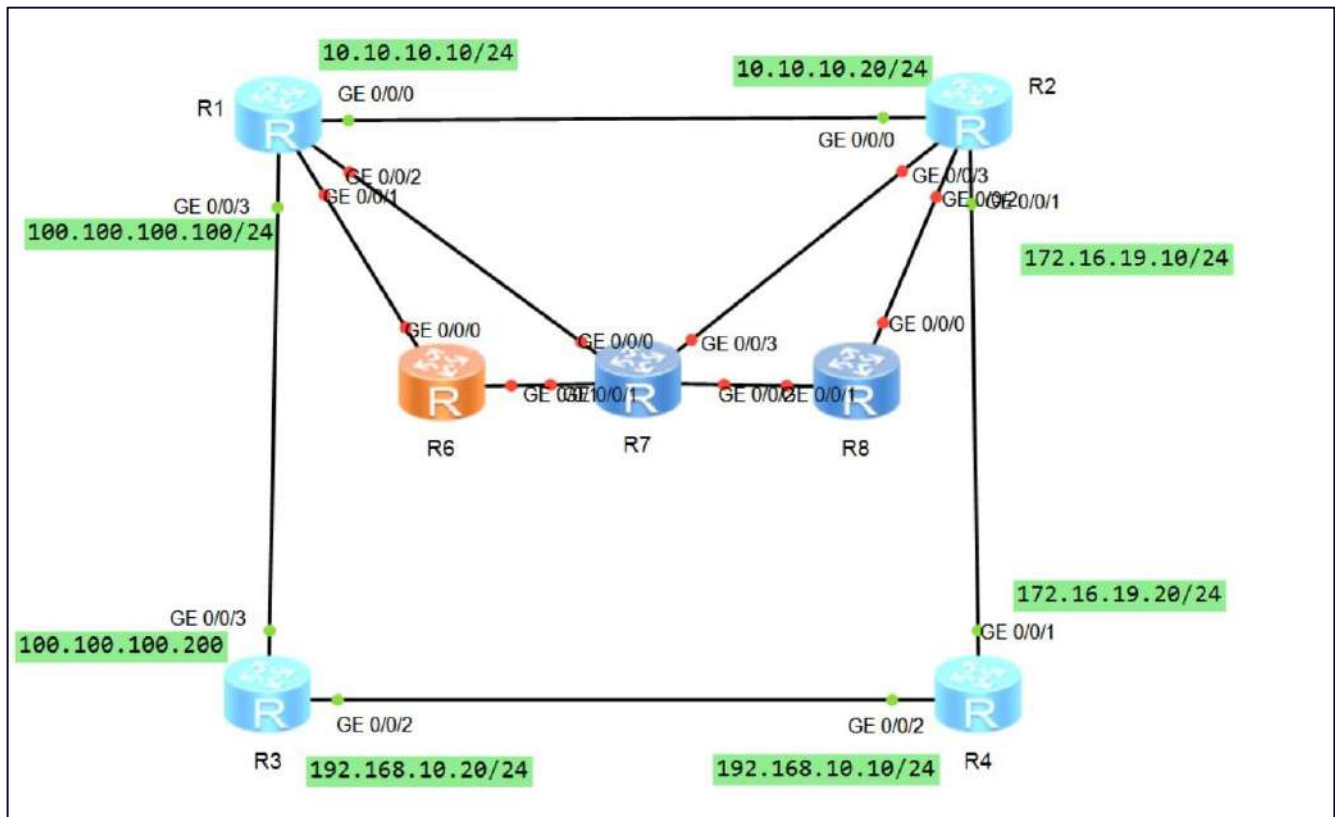
NED University

Name: MUHAMMAD OBAID

Roll Number : CT-23025 (Section A)



Submit the configured topology such that every router should be accessible to every router and also include backup routes on routers as well using Static Routes.



Configuration of R1:

```
[R1]interface G
[R1]interface GigabitEthernet 0/0/0
[R1-GigabitEthernet0/0/0]ip address 10.10.10.10 24
[R1-GigabitEthernet0/0/0]quit
[R1]interface GigabitEthernet 0/0/1
[R1-GigabitEthernet0/0/1]ip address 10.1.5.1 24
[R1-GigabitEthernet0/0/1]quit
[R1]interface GigabitEthernet 0/0/2
[R1-GigabitEthernet0/0/2]ip address 10.1.6.1 24
[R1-GigabitEthernet0/0/2]quit
[R1]interface GigabitEthernet 0/0/3
[R1-GigabitEthernet0/0/3]ip address 100.100.100.100 24
[R1-GigabitEthernet0/0/3]quit
[R1]display ip interface brief
```

```
[R1]ip route-static 172.16.19.0 24 10.10.10.20
[R1]ip route-static 172.16.19.0 24 10.2.6.2 preference 70
[R1]ip route-static 10.2.6.0 24 10.1.6.6
[R1]ip route-static 10.2.6.0 24 10.10.10.20 preference 70
[R1]ip route-static 10.2.7.0 24 10.10.10.20
[R1]ip route-static 10.2.7.0 24 10.1.6.6 preference 70
[R1]ip route-static 192.168.10.0 24 100.100.100.200
[R1]ip route-static 192.168.10.0 24 10.10.10.20 preference 70
[R1]ip route-static 10.6.7.0 24 10.1.6.6
[R1]ip route-static 10.6.7.0 24 10.10.10.20 preference 70
[R1]ip route-static 10.5.6.0 24 10.1.5.5
[R1]ip route-static 10.5.6.0 24 10.1.6.6 preference 70
[R1]display ip routing-table
```

For R2:

```
<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2]interface G
[R2]interface GigabitEthernet 0/0/0
[R2-GigabitEthernet0/0/0]ip address 10.10.10.20 24
[R2-GigabitEthernet0/0/0]quit
[R2]interface GigabitEthernet 0/0/1
[R2-GigabitEthernet0/0/1]ip address 172.16.19.10 24
[R2-GigabitEthernet0/0/1]quit
[R2]interface GigabitEthernet 0/0/2
[R2-GigabitEthernet0/0/2]ip address 10.2.7.2 24
[R2-GigabitEthernet0/0/2]quit
[R2]interface GigabitEthernet 0/0/3
[R2-GigabitEthernet0/0/3]ip address 10.2.6.2 24
[R2-GigabitEthernet0/0/3]quit
[R2]display ip interface brief

[R2]ip route-static 100.100.100.0 24 10.10.10.10
[R2]ip route-static 100.100.100.0 24 10.2.6.6 preference 70
[R2]ip route-static 10.1.6.0 24 10.2.6.6
[R2]ip route-static 10.1.6.0 24 10.10.10.10 preference 70
[R2]ip route-static 10.1.5.0 24 10.10.10.10
[R2]ip route-static 10.1.5.0 24 10.2.6.6 preference 70
[R2]ip route-static 192.168.10.0 24 172.16.19.20
[R2]ip route-static 192.168.10.0 24 10.10.10.10 preference 70
[R2]ip route-static 10.5.6.0 24 10.2.6.6
[R2]ip route-static 10.5.6.0 24 10.10.10.10 preference 70
[R2]ip route-static 10.6.7.0 24 10.2.7.7
[R2]ip route-static 10.6.7.0 24 10.2.6.6 preference 70
[R2]display ip routing-table
```

For R3:

```

R3
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R3
[R3]interface G
[R3]interface GigabitEthernet 0/0/2
[R3-GigabitEthernet0/0/2]ip address 192.168.10.20 24
[R3-GigabitEthernet0/0/2]quit
[R3]interface GigabitEthernet 0/0/3
[R3-GigabitEthernet0/0/3]ip address 100.100.100.200 24
[R3-GigabitEthernet0/0/3]quit
[R3]ip route 0.0.0.0 0 100.100.100.100
      ^
Error: Wrong parameter found at '^' position.
[R3]ip route 0.0.0.0 0.0.0.0 100.100.100.100
      ^
Error: Wrong parameter found at '^' position.
[R3]ip route-static 0.0.0.0 0 100.100.100.100
[R3]ip route-static 0.0.0.0 0 192.168.10.10 preference 70
[R3]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
  Destinations : 7          Routes : 7

Destination/Mask    Proto   Pre  Cost   Flags NextHop         Interface
-----
0.0.0.0/0           Static  60   0       RD   100.100.100.100 GigabitEthernet
0/0/3
100.100.100.0/24    Direct  0     0       D   100.100.100.200 GigabitEthernet
0/0/3
100.100.100.200/32  Direct  0     0       D   127.0.0.1       GigabitEthernet
0/0/3
127.0.0.0/8         Direct  0     0       D   127.0.0.1       InLoopBack0
127.0.0.1/32        Direct  0     0       D   127.0.0.1       InLoopBack0
192.168.10.0/24     Direct  0     0       D   192.168.10.20   GigabitEthernet
0/0/2
192.168.10.20/32    Direct  0     0       D   127.0.0.1       GigabitEthernet
0/0/2

[R3]quit
<R3>save

```

For R4:

```

<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R4
[R4]interface G
[R4]interface GigabitEthernet 0/0/1
[R4-GigabitEthernet0/0/1]ip address 172.16.19.20 24
[R4-GigabitEthernet0/0/1]quit
[R4]interface GigabitEthernet 0/0/2
[R4-GigabitEthernet0/0/2]ip address 192.168.10.10 24
[R4-GigabitEthernet0/0/2]quit
[R4]ip route-static 0.0.0.0 0.0.0.0 172.16.19.10
[R4]ip route-static 0.0.0.0 0.0.0.0 192.168.10.20 preference 70
[R4]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
  Destinations : 7          Routes : 7

Destination/Mask    Proto   Pre  Cost   Flags NextHop         Interface
-----
0.0.0.0/0           Static  60   0       RD   172.16.19.10    GigabitEthernet
0/0/1
127.0.0.0/8         Direct  0     0       D   127.0.0.1       InLoopBack0
127.0.0.1/32        Direct  0     0       D   127.0.0.1       InLoopBack0
172.16.19.0/24      Direct  0     0       D   172.16.19.20    GigabitEthernet
0/0/1
172.16.19.20/32     Direct  0     0       D   127.0.0.1       GigabitEthernet
0/0/1
192.168.10.0/24     Direct  0     0       D   192.168.10.10   GigabitEthernet
0/0/2
192.168.10.10/32    Direct  0     0       D   127.0.0.1       GigabitEthernet
0/0/2

[R4]quit
<R4>save

```

For R5:

```
R5
The device is running!

<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R5
[R5]interface G
[R5]interface GigabitEthernet 0/0/0
[R5-GigabitEthernet0/0/0]ip address 10.1.5.5 24
[R5-GigabitEthernet0/0/0]quit
[R5]interface GigabitEthernet 0/0/1
[R5-GigabitEthernet0/0/1]ip address 10.5.6.5 24
[R5-GigabitEthernet0/0/1]quit
[R5]ip route-static 0.0.0.0 0 10.1.5.1
[R5]ip route-static 0.0.0.0 0 10.5.6.6 preference 70
[R5]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
  Destinations : 7          Routes : 7

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface
-----
0/0/0              0.0.0.0/0   Static 60    0          RD  10.1.5.1       GigabitEthernet
0/0/0              10.1.5.0/24 Direct  0     0          D   10.1.5.5       GigabitEthernet
0/0/0              10.1.5.5/32 Direct  0     0          D   127.0.0.1      GigabitEthernet
0/0/0              10.5.6.0/24 Direct  0     0          D   10.5.6.5       GigabitEthernet
0/0/1              10.5.6.5/32 Direct  0     0          D   127.0.0.1      GigabitEthernet
0/0/1              127.0.0.0/8 Direct  0     0          D   127.0.0.1      InLoopBack0
                  127.0.0.1/32 Direct  0     0          D   127.0.0.1      InLoopBack0

[R5]quit
<R5>save
```

For R7:

```
<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R7
[R7]interface G
[R7]interface GigabitEthernet 0/0/0
[R7-GigabitEthernet0/0/0]ip address 10.2.7.7 24
[R7-GigabitEthernet0/0/0]quit
[R7]interface GigabitEthernet 0/0/1
[R7-GigabitEthernet0/0/1]ip address 10.6.7.7 24
[R7-GigabitEthernet0/0/1]quit
[R7]ip route-static 0.0.0.0 0 10.2.7.2
[R7]ip route-static 0.0.0.0 0 10.6.7.6 preference 70
[R7]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
  Destinations : 7          Routes : 7

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface
-----
0/0/0              0.0.0.0/0   Static 60    0          RD  10.2.7.2       GigabitEthernet
0/0/0              10.2.7.0/24 Direct  0     0          D   10.2.7.7       GigabitEthernet
0/0/0              10.2.7.7/32 Direct  0     0          D   127.0.0.1      GigabitEthernet
0/0/0              10.6.7.0/24 Direct  0     0          D   10.6.7.7       GigabitEthernet
0/0/1              10.6.7.7/32 Direct  0     0          D   127.0.0.1      GigabitEthernet
0/0/1              127.0.0.0/8 Direct  0     0          D   127.0.0.1      InLoopBack0
                  127.0.0.1/32 Direct  0     0          D   127.0.0.1      InLoopBack0

[R7]quit
<R7>save
```

For R6:

```
<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R6
[R6]interface G
[R6]interface GigabitEthernet 0/0/0
[R6-GigabitEthernet0/0/0]ip address 10.1.6.6 24
[R6-GigabitEthernet0/0/0]quit
[R6]interface GigabitEthernet 0/0/1
[R6-GigabitEthernet0/0/1]ip address 10.5.6.6 24
[R6-GigabitEthernet0/0/1]quit
[R6]interface GigabitEthernet 0/0/2
[R6-GigabitEthernet0/0/2]ip address 10.6.7.6 24
[R6-GigabitEthernet0/0/2]quit
[R6]interface GigabitEthernet 0/0/3
[R6-GigabitEthernet0/0/3]ip address 10.2.6.6 24
[R6-GigabitEthernet0/0/3]quit
```

```
[R6]ip route-static 100.100.100.0 24 10.1.6.1
[R6]ip route-static 100.100.100.0 24 10.1.6.1
Feb 21 2026 15:05:16-08:00 R6 DS/4/DATASYNC_CFGCHANGE:OID 1.3.6.1.4.1.2011.5.25.
191.3.1 configurations have been changed. The current change number is 6, the ch
ange loop count is 0, and the maximum number of records is 4095.
Error: The route already exists.
[R6]ip route-static 100.100.100.0 24 10.2.6.2 preference 70
[R6]ip route-static 100.100.100.0 24 10.2.6.2 preference 70
Feb 21 2026 15:05:56-08:00 R6 DS/4/DATASYNC_CFGCHANGE:OID 1.3.6.1.4.1.2011.5.25.
191.3.1 configurations have been changed. The current change number is 7, the ch
ange loop count is 0, and the maximum number of records is 4095.
Error: The route already exists.
[R6]quit
<R6>u t m
Info: Current terminal monitor is off.
<R6>system-view
Enter system view, return user view with Ctrl+Z.
[R6]ip route-static 10.1.5.0 24 10.5.6.5
[R6]ip route-static 10.1.5.0 24 10.1.6.1 preference 70
[R6]ip route-static 172.16.19.0 24 10.2.6.2
[R6]ip route-static 172.16.19.0 24 10.1.6.1 preference 70
[R6]ip route-static 10.2.7.0 24 10.6.7.7
[R6]ip route-static 10.2.7.0 24 10.2.6.2 preference 70
[R6]ip route-static 10.10.10.0 24 10.1.6.1
[R6]ip route-static 10.10.10.0 24 10.2.6.2 preference 70
[R6]ip route-static 192.168.10.0 24 10.1.6.1
[R6]ip route-static 192.168.10.0 24 10.2.6.2 prefernece 70
^
Error:Too many parameters found at '^' position.
[R6]ip route-static 192.168.10.0 24 10.2.6.2 preference 70
[R6]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
Destinations : 16 Routes : 16

Destination/Mask Proto Pre Cost Flags NextHop Interface
-----
10.1.5.0/24 Static 60 0 RD 10.5.6.5 GigabitEthernet
```

Lab 9 (CCN)

WEDNESDAY, APRIL 22, 2026

NED University

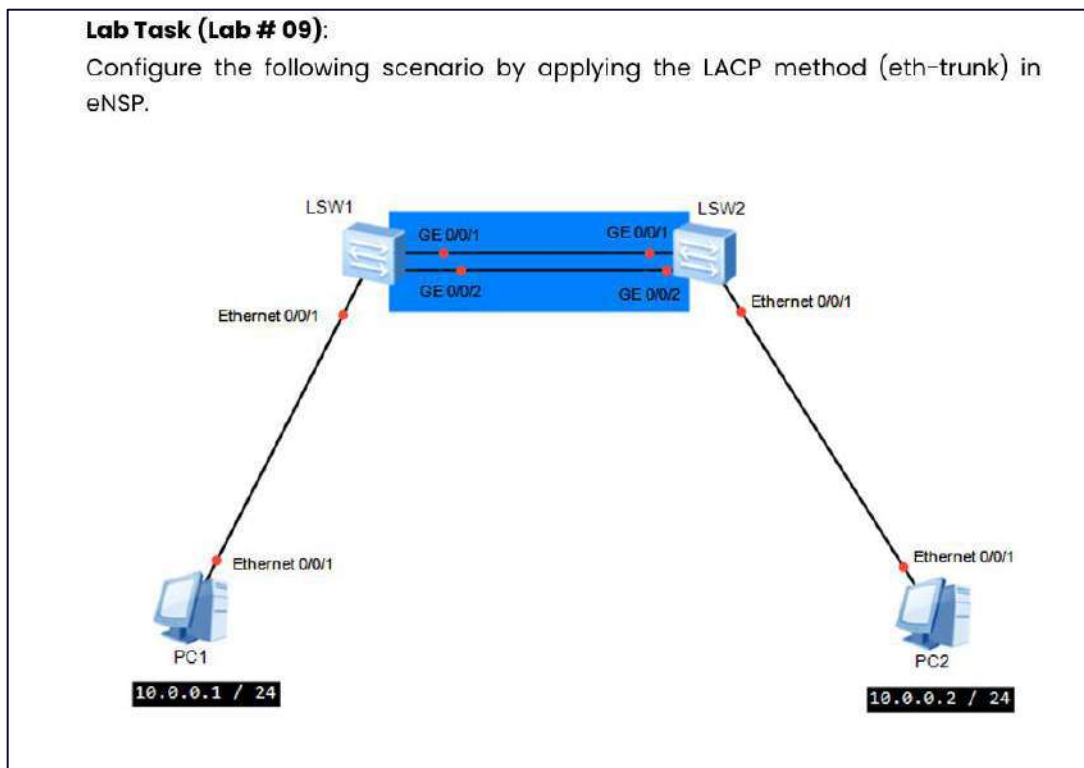
Name: MUHAMMAD OBAID

Roll Number : CT-23025 (Section A)

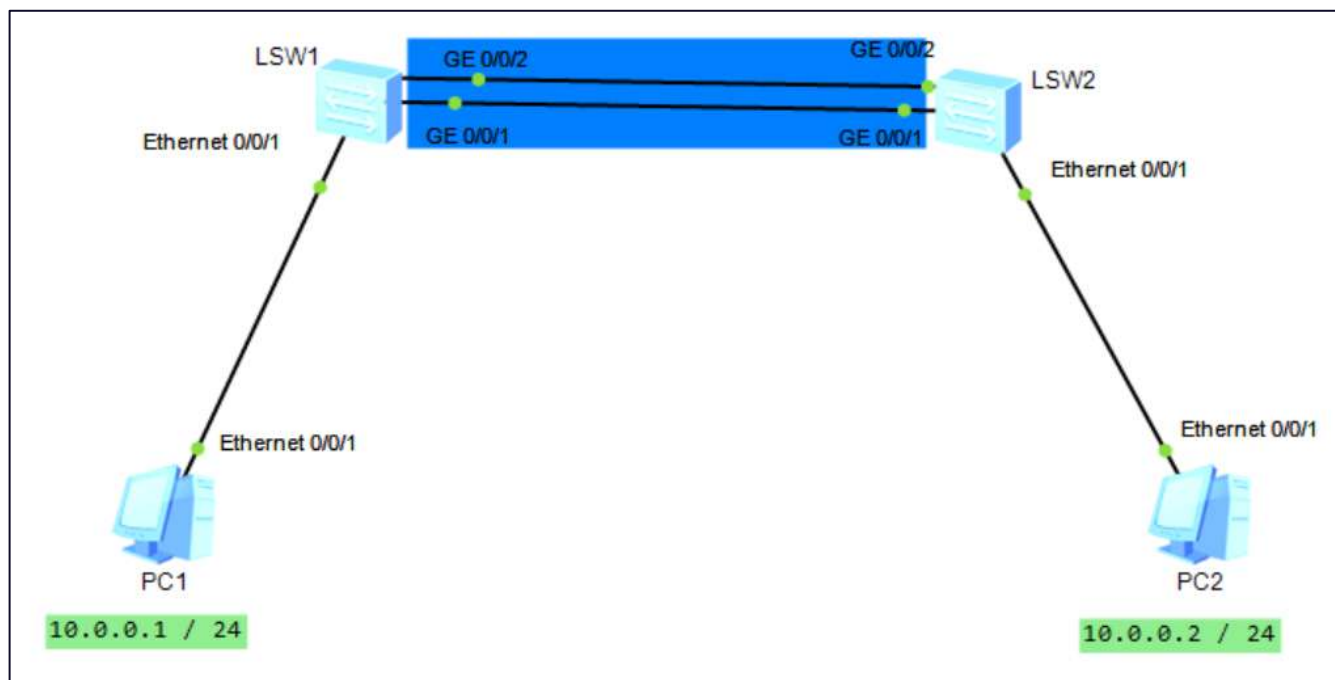


Lab Task (Lab # 09) :

Configure the following scenario by applying the LACP method (eth-trunk) in eNSP.



Configuration:



Switch 1:

```
LSW1
<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname LSW1
[LSW1]interface Eth
[LSW1]interface Ethernet 0/0/1
[LSW1-Ethernet0/0/1]port link-type access
[LSW1-Ethernet0/0/1]quit
[LSW1]interface Eth-Trunk 1
[LSW1-Eth-Trunk1]trunkport Gig
[LSW1-Eth-Trunk1]trunkport GigabitEthernet 0/0/1 to 0/0/2
Info: This operation may take a few seconds. Please wait for a moment...done.
[LSW1-Eth-Trunk1]mode lacp-static
Error: Error in changing trunk working mode. There is(are) port(s) in the trunk.

[LSW1-Eth-Trunk1]undo trunkport GigabitEthernet 0/0/1 to 0/0/2
Info: This operation may take a few seconds. Please wait for a moment...done.
[LSW1-Eth-Trunk1]mode lacp-static
[LSW1-Eth-Trunk1]trunkport GigabitEthernet 0/0/1 to 0/0/2
Info: This operation may take a few seconds. Please wait for a moment...done.
[LSW1-Eth-Trunk1]port link-type trunk
[LSW1-Eth-Trunk1]port trunk allow-pass vlan all
[LSW1-Eth-Trunk1]quit
[LSW1]quit
<LSW1>save
The current configuration will be written to the device.
Are you sure to continue?[Y/N]Y
Info: Please input the file name ( *.cfg, *.zip ) [vrpcfg.zip]:
Now saving the current configuration to the slot 0.
Save the configuration successfully.
<LSW1>sysate
<LSW1>system-view
Enter system view, return user view with Ctrl+Z.
[LSW1]interface Gig
[LSW1]interface GigabitEthernet 0/0/1
[LSW1-GigabitEthernet0/0/1]shutdown
[LSW1-GigabitEthernet0/0/1]quit
[LSW1]display eth-trunk 1
Eth-Trunk1's state information is:
Local:
LAG ID: 1                WorkingMode: STATIC
Preempt Delay: Disabled   Hash arithmetic: According to SIP-XOR-DIP
System Priority: 32768     System ID: 4clif-ccda-71d3
Least Active-linknumber: 1 Max Active-linknumber: 8
Operate status: up        Number Of Up Port In Trunk: 1
-----
ActorPortName      Status   PortType PortPri PortNo PortKey PortState Weight
GigabitEthernet0/0/1 Unselect 1GE      32768   2      305    10100010 1
GigabitEthernet0/0/2 Selected 1GE      32768   3      305    10111100 1
-----
Partner:
-----
ActorPortName      SysPri   SystemID      PortPri PortNo PortKey PortState
GigabitEthernet0/0/1 0         0000-0000-0000 0        0      0      10100011
GigabitEthernet0/0/2 32768    4clif-cc27-2be4 32768    3      305    10111100
-----
[LSW1]

[LSW1]interface GigabitEthernet 0/0/1
[LSW1-GigabitEthernet0/0/1]undo shutdown
[LSW1-GigabitEthernet0/0/1]quit
[LSW1]quit
<LSW1>save
The current configuration will be written to the device.
Are you sure to continue?[Y/N]Y
Now saving the current configuration to the slot 0.
Save the configuration successfully.
<LSW1>
<LSW1>
```

Switch 2:

```

LSW2
The device is running!
<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>sys
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname LSW2
[LSW2]interface Ethe
[LSW2]interface Ethernet 0/0/1
[LSW2-Ethernet0/0/1]port link-type access
[LSW2-Ethernet0/0/1]quit
[LSW2]interface Eth-Tr
[LSW2]interface Eth-Trunk 1
[LSW2-Eth-Trunk1]mode lacp-static
[LSW2-Eth-Trunk1]trunkport Gig
[LSW2-Eth-Trunk1]trunkport GigabitEthernet 0/0/1 to 0/0/2
Info: This operation may take a few seconds. Please wait for a moment...done.
[LSW2-Eth-Trunk1]port link-type trunk
[LSW2-Eth-Trunk1]port trunk allow-pass vlan all
[LSW2-Eth-Trunk1]quit
[LSW2]display eth-trunk 1
Eth-Trunk1's state information is:
Local:
LAG ID: 1                WorkingMode: STATIC
Preempt Delay: Disabled   Hash arithmetic: According to SIP-XOR-DIP
System Priority: 32768     System ID: 4c1f-cc27-2be4
Least Active-linknumber: 1 Max Active-linknumber: 8
Operate status: up        Number Of Up Port In Trunk: 2
-----
ActorPortName      Status  PortType  PortPri  PortNo  PortKey  PortState  Weight
GigabitEthernet0/0/1  Selected 1GE      32768    2       305     10111100   1
GigabitEthernet0/0/2  Selected 1GE      32768    3       305     10111100   1

Partner:
-----
ActorPortName      SysPri  SystemID      PortPri  PortNo  PortKey  PortState
GigabitEthernet0/0/1  32768   4c1f-ccda-71d3 32768    2       305     10111100
GigabitEthernet0/0/2  32768   4c1f-ccda-71d3 32768    3       305     10111100

[LSW2]quit
<LSW2>save
The current configuration will be written to the device.
Are you sure to continue?[Y/N]Y
Info: Please input the file name ( *.cfg, *.zip ) [vrpcfg.zip]:
Now saving the current configuration to the slot 0.
Save the configuration successfully.
<LSW2>sys
<LSW2>system-view
Enter system view, return user view with Ctrl+Z.
[LSW2]display eth-trunk 1
Eth-Trunk1's state information is:
Local:
LAG ID: 1                WorkingMode: STATIC
Preempt Delay: Disabled   Hash arithmetic: According to SIP-XOR-DIP
System Priority: 32768     System ID: 4c1f-cc27-2be4
Least Active-linknumber: 1 Max Active-linknumber: 8
Operate status: up        Number Of Up Port In Trunk: 1
-----
ActorPortName      Status  PortType  PortPri  PortNo  PortKey  PortState  Weight
GigabitEthernet0/0/1  Unselect 1GE      32768    2       305     10100010   1
GigabitEthernet0/0/2  Selected 1GE      32768    3       305     10111100   1

Partner:
-----
ActorPortName      SysPri  SystemID      PortPri  PortNo  PortKey  PortState
GigabitEthernet0/0/1  0       0000-0000-0000 0       0       0       10100011
GigabitEthernet0/0/2  32768   4c1f-ccda-71d3 32768    3       305     10111100

[LSW2]

```

For manual mode, can do explicitly also:

- # Note: In manual mode, we do not use the 'mode' command
- # or we set it explicitly if supported. By default, Eth-Trunk is manual.
- [LSW1-Eth-Trunk1] undo mode lacp-static
- [LSW1-Eth-Trunk1] mode manual load-balance

Feature	LACP (Static)	Manual Mode
Negotiation	Switches exchange LACPDU packets to verify the link.	No protocol exchange; ports are bundled immediately.
Fault Detection	Detects if a cable is plugged into the wrong port or switch.	Cannot detect misconfigurations (can lead to loops).

Reliability	Highly recommended for production environments.	Usually used when the peer device doesn't support LACP.
--------------------	---	---

Ping:

With both interfaces UP/Selected:

```

PC1
Basic Config  Command  MCPacket  UdpPacket  Console
Welcome to use PC Simulator!

PC>ping 10.0.0.2

Ping 10.0.0.2: 32 data bytes, Press Ctrl_C to break
From 10.0.0.2: bytes=32 seq=1 ttl=128 time=94 ms
From 10.0.0.2: bytes=32 seq=2 ttl=128 time=78 ms
From 10.0.0.2: bytes=32 seq=3 ttl=128 time=62 ms
From 10.0.0.2: bytes=32 seq=4 ttl=128 time=78 ms
From 10.0.0.2: bytes=32 seq=5 ttl=128 time=78 ms

--- 10.0.0.2 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 62/78/94 ms

PC>display arp -a
Invalid command!

PC>arp -a

Internet Address    Physical Address    Type
10.0.0.2            54-89-98-22-20-FE  dynamic

```

With interface Gig 0/0/1 Down/Unselected (due to being shutdown). Since initially ARP-cache has the dest-route via next hp of Gig 0/0/1 interface but since it was down so was giving “Request Timed Out”. The timeout was the "gap" while the switches and PCs updated their internal maps (ARP/MAC tables) to realize that GE 0/0/2 was the only remaining valid path.

Then when cache was reset, it again successfully pinged via Gig 0/0/2

```

PC>ping 10.0.0.2

Ping 10.0.0.2: 32 data bytes, Press Ctrl_C to break
Request timeout!
Request timeout!
Request timeout!
Request timeout!
Request timeout!

--- 10.0.0.2 ping statistics ---
 5 packet(s) transmitted
 0 packet(s) received
100.00% packet loss

PC>arp -a

Internet Address    Physical Address    Type

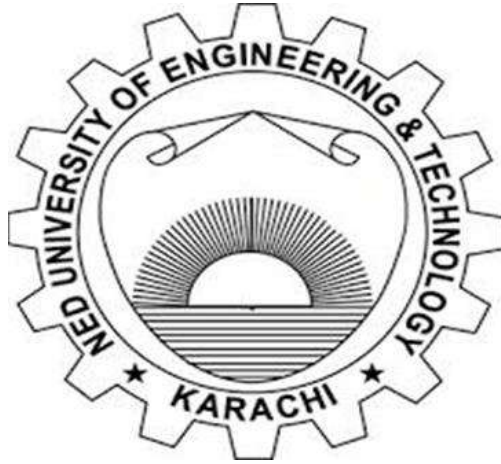
PC>ping 10.0.0.2

Ping 10.0.0.2: 32 data bytes, Press Ctrl_C to break
From 10.0.0.2: bytes=32 seq=1 ttl=128 time=62 ms
From 10.0.0.2: bytes=32 seq=2 ttl=128 time=47 ms
From 10.0.0.2: bytes=32 seq=3 ttl=128 time=109 ms
From 10.0.0.2: bytes=32 seq=4 ttl=128 time=62 ms
From 10.0.0.2: bytes=32 seq=5 ttl=128 time=78 ms

--- 10.0.0.2 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 47/71/109 ms

PC>

```



COMPUTER COMMUNICATION NETWORKS

(CT-376)

LAB INSTRUCTOR: Mr. Ahmed Zaki Shaikh

Lab Session # 10

Name: Muhammad Obaid

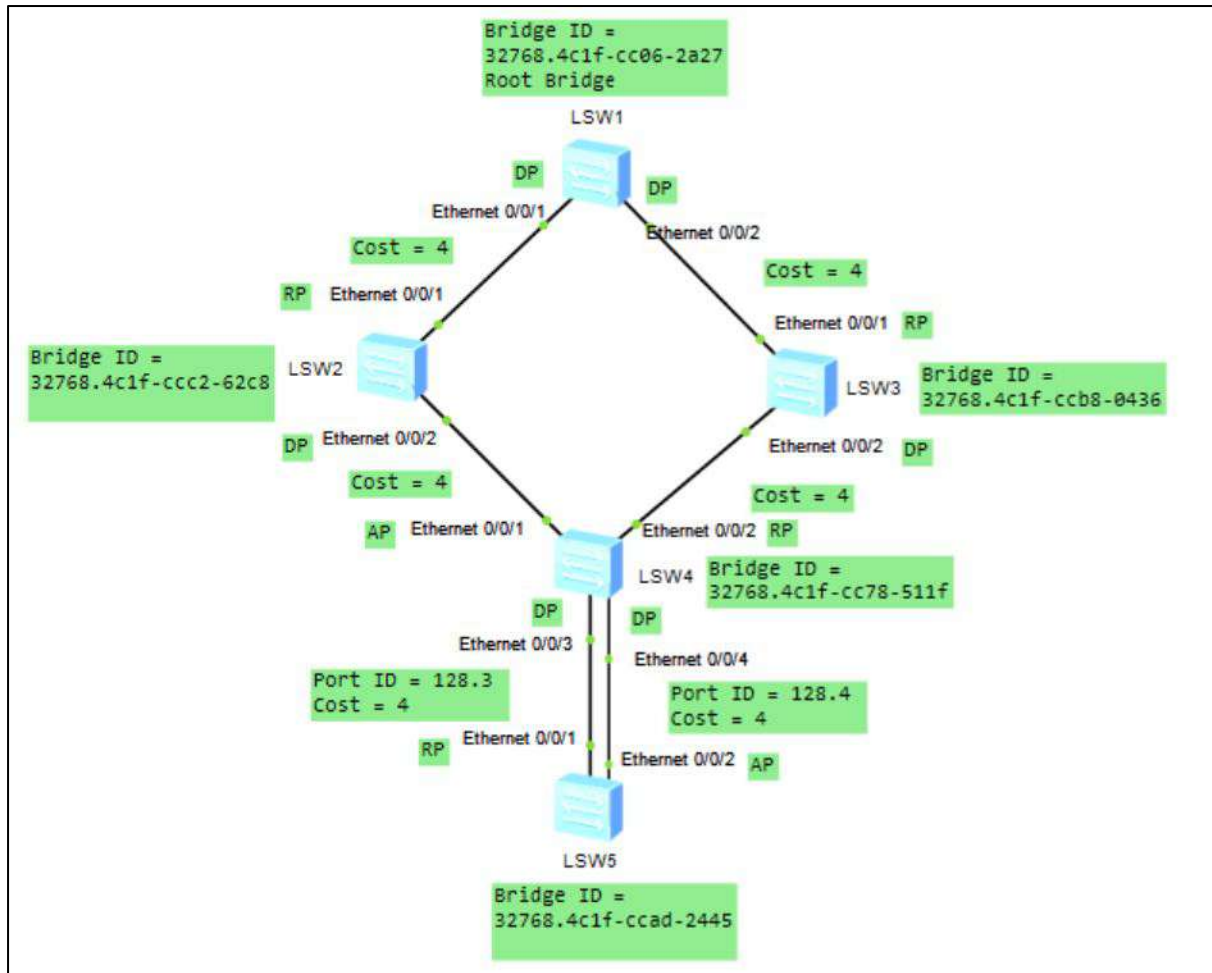
Class: TSCS Sec (A)

Batch: 2023

Roll Number: CT-23025

Q. State reason and identify the root bridge, root port (RP), designated port (DP) and alternative port (or blocked port), if STP is applied in the following scenarios:

Scenario # 1:



```

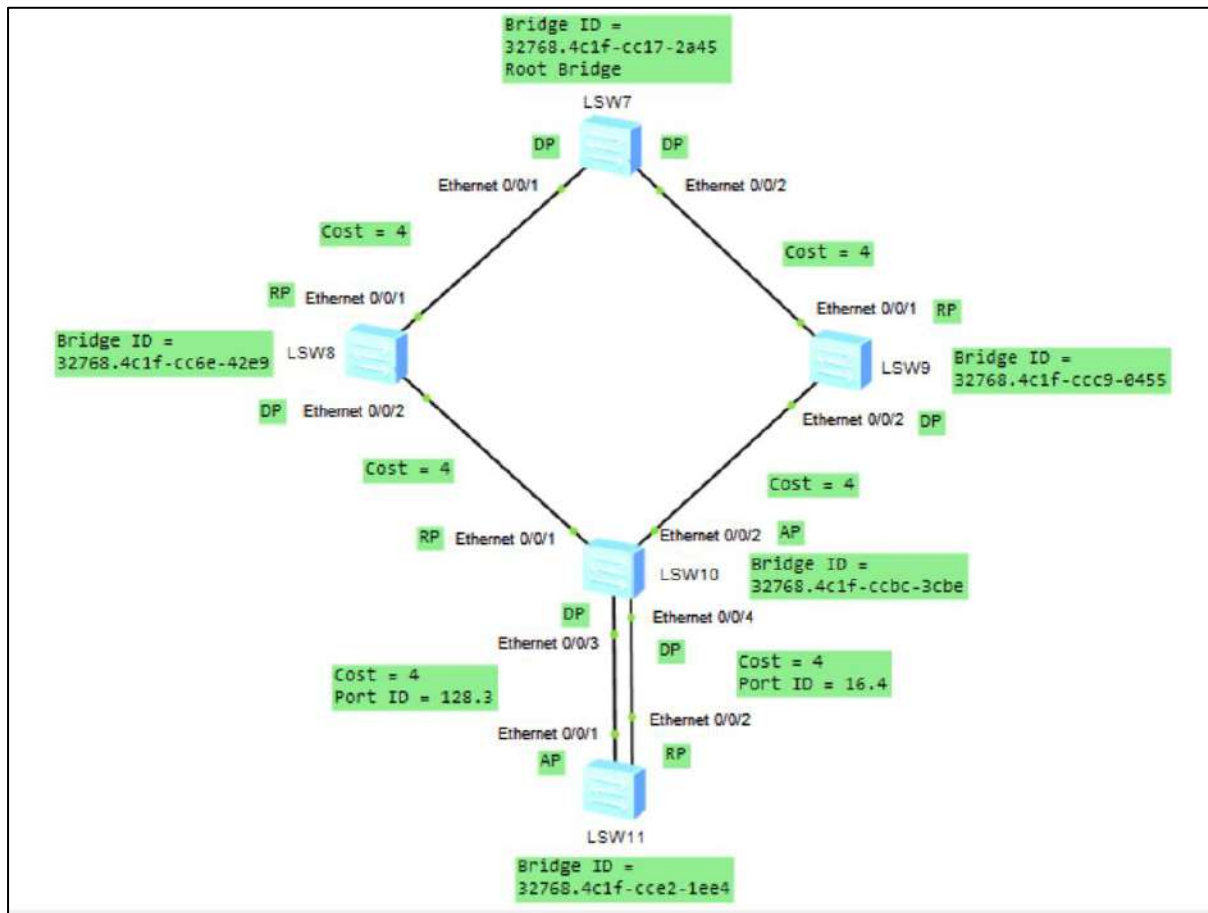
[LSW1]interface Ethernet 0/0/1
[LSW1-Ethernet0/0/1]stp cost 4
[LSW1-Ethernet0/0/1]quit
[LSW1]interface Ethernet 0/0/2
[LSW1-Ethernet0/0/2]stp cost 4
[LSW1-Ethernet0/0/2]quit
[LSW1]quit

```

Reasons:

- LSW1 having the lowest Bridge ID is the Root Bridge (both Designated Ports)
- LSW2 and LSW3 Ethernet 0/0/1 having lowest cost 4 to Root Bridge (Root Port)
- LSW4 having tie at cost 8 on both Ethernet 0/0/1 and Ethernet 0/0/2
 - LSW3 has the lowest Bridge ID compared to LSW2
 - LSW4 Ethernet 0/0/2 being the Root Port
 - Opposite to LSW4 has LSW3 Ethernet 0/0/2 will be Designated Port
- LSW2 Ethernet 0/0/2 has low cost compared to LSW4 Ethernet 0/0/1
 - LSW2 Ethernet 0/0/2 will be Designated Port
 - LSW4 Ethernet 0/0/1 will be Alternate Port
- LSW5 having tie at cost 12 on both Ethernet 0/0/1 and Ethernet 0/0/2
 - Cost and Bridge ID is same
 - Checking opposite port number, LSW4 Ethernet 0/0/3 is less than Ethernet 0/0/4
 - LSW5 Ethernet 0/0/1 will be Root Port
 - Opposite of Root Port, LSW4 Ethernet 0/0/3 will be Designated Port
- LSW4 Ethernet 0/0/4 has less cost to Root Bridge than LSW5 Ethernet 0/0/2
 - LSW4 Ethernet 0/0/4 will be Designated Port
 - LSW5 Ethernet 0/0/2 will be Alternate Port

Scenario # 2:



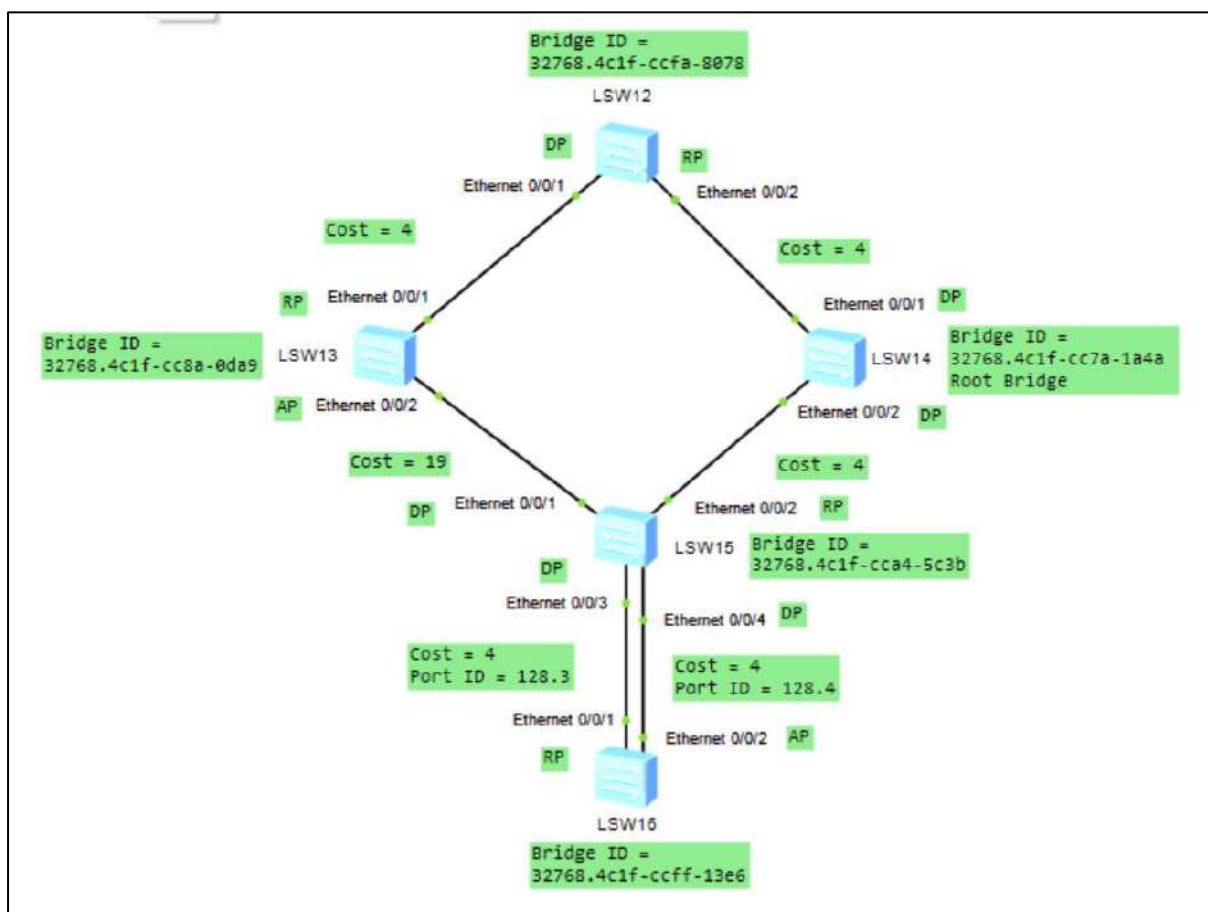
Specify Cost and Port Priority:

```
[LSW10]interface Ethernet 0/0/1
[LSW10-Ethernet0/0/1]stp cost 4
[LSW10-Ethernet0/0/1]quit
[LSW10]interface Ethernet 0/0/2
[LSW10-Ethernet0/0/2]stp cost 4
[LSW10-Ethernet0/0/2]quit
[LSW10]interface Ethernet 0/0/3
[LSW10-Ethernet0/0/3]stp cost 4
[LSW10-Ethernet0/0/3]stp port priority 128
[LSW10-Ethernet0/0/3]quit
[LSW10]interface Ethernet 0/0/4
[LSW10-Ethernet0/0/4]stp cost 4
[LSW10-Ethernet0/0/4]stp port priority 16
[LSW10-Ethernet0/0/4]quit
[LSW10]quit
```

Reasons:

- LSW11 having cost tie on both Ethernet 0/0/1 and Ethernet 0/0/2
 - Ethernet 0/0/2 has low Port ID than Ethernet 0/0/1
 - Ethernet 0/0/2 will be Root Port
 - Opposite of LSW11 Ethernet 0/0/2 will be Designated Port (LSW10 Ethernet 0/0/4)
- LSW10 Ethernet 0/0/3 having low cost than LSW11 Ethernet 0/0/1
 - LSW10 Ethernet 0/0/3 will be Designated Port
 - LSW11 Ethernet 0/0/1 will be Alternate Port
- Rest of the ports of Switches will be same as above scenario # 1

Scenario # 3:



Reasons:

- LSW14 has the lowest Bridge ID (Root Bridge)
 - Both ports of LSW14 Ethernet 0/0/1 and Ethernet 0/0/2 will be Designated Ports
- LSW12 and LSW15 both ports Ethernet 0/0/2 will be Root Ports having less cost 4 to reach Root Bridge
- LSW13 Ethernet 0/0/1 has less cost 8 than Ethernet 0/0/2 23
 - Ethernet 0/0/1 will be Root Port

- Opposite LSW12 Ethernet 0/0/1 will be Designated Port
- LSW15 Ethernet 0/0/1 has less cost 4 than LSW13 Ethernet 0/0/2 23
 - LSW15 Ethernet 0/0/1 will be Designated Port
 - LSW13 Ethernet 0/0/2 will be Alternate Port
- LSW16 have tie cost 8 on both Ethernet 0/0/1 and Ethernet 0/0/2
 - Cost and Bridge ID is same
 - Ethernet 0/0/1 Port ID is less than Ethernet 0/0/2 Port ID
 - Ethernet 0/0/1 will be Root Port
 - Opposite LSW15 Ethernet 0/0/3 will be Designated Port
- LSW15 Ethernet 0/0/4 has less cost 4 than LSW16 Ethernet 0/0/2 8
 - LSW15 Ethernet 0/0/4 will be Designated Port
 - LSW16 Ethernet 0/0/2 will be Alternate Port

Lab 11 (CCN)

WEDNESDAY, MAY 6, 2026

NED University

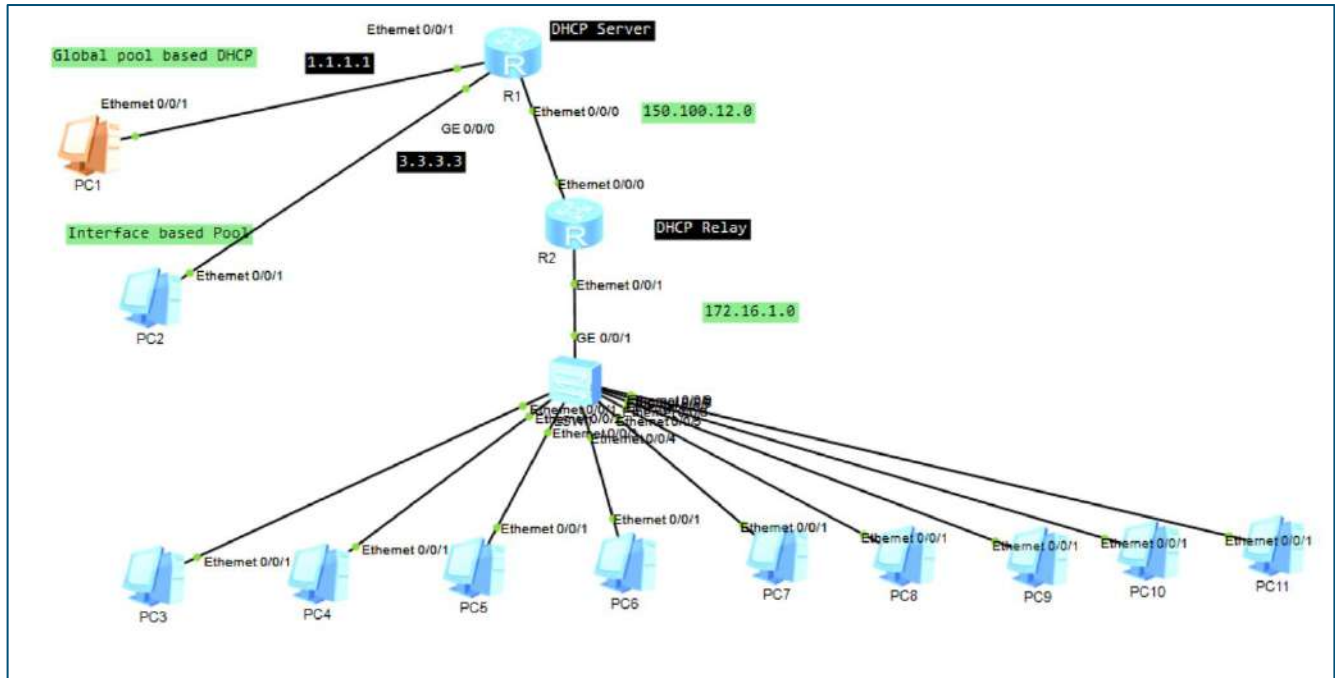
Name: MUHAMMAD OBAID

Roll Number : CT-23025 (Section A)



Lab Task (Lab # 11) :

Implement the following scenario by implementing the global pool and interface based DHCP server along with frame relay DHCP server.



Configuration:

```
R1
The device is running!

<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>sysname R1
^
Error: Unrecognized command found at '^' position.
<Huawei>sys
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]dhcp enable global
^
Error:Too many parameters found at '^' position.
[R1]dhcp enable
Info: The operation may take a few seconds. Please wait for a moment.done.
[R1]ip pool pool_pcl
Info:It's successful to create an IP address pool.
[R1-ip-pool-pool_pcl]network 1.1.1.0 mask 255.255.255.0
[R1-ip-pool-pool_pcl]gateway-list 1.1.1.1
[R1-ip-pool-pool_pcl]q
[R1]interface Eth
[R1]interface Ethernet 0/0/1
[R1-Ethernet0/0/1]ip address 1.1.1.1 24
[R1-Ethernet0/0/1]dhcp select global
[R1-Ethernet0/0/1]q
[R1]interface Gig
[R1]interface GigabitEthernet 0/0/0
[R1-GigabitEthernet0/0/0]ip address 3.3.3.3 24
[R1-GigabitEthernet0/0/0]dhcp select interface
[R1-GigabitEthernet0/0/0]q
[R1]interface Et
[R1]interface Ethe
[R1]interface Ethernet 0/0/0
[R1-Ethernet0/0/0]ip address 150.100.12.1 24
[R1-Ethernet0/0/0]q
[R1]ip pool pool_relay
Info:It's successful to create an IP address pool.
[R1-ip-pool-pool_relay]network 172.16.1.0 mask 255.255.255.0
[R1-ip-pool-pool_relay]gateway-list 172.16.1.1
```

```

[R1-ip-pool-pool_relay]q
[R1]ip route-static 172.16.1.0 24 150.100.12.2
[R1]q
<R1>save
The current configuration will be written to the device.
Are you sure to continue?[Y/N]Y
Info: Please input the file name ( *.cfg, *.zip ) [vrpcfg.zip]:
Now saving the current configuration to the slot 17.
Save the configuration successfully.
<R1> User interface con0 is available

Please Press ENTER.

<R1>sys
Enter system view, return user view with Ctrl+Z.
[R1]interface Eth
[R1]interface Ethernet 0/0/0
[R1-Ethernet0/0/0]dhcp select global
[R1-Ethernet0/0/0]q
May 6 2026 22:10:59-08:00 R1 DS/4/DATASYNC_CFGCHANGE:OID 1.3.6.1.4.1.2011.5.25.
191.3.1 configurations have been changed. The current change number is 15, the c
hange loop count is 0, and the maximum number of records is 4095.
[R1]q
<R1>save
The current configuration will be written to the device.
Are you sure to continue?[Y/N]Y
Now saving the current configuration to the slot 17.
May 6 2026 22:11:05-08:00 R1 %%01CFM/4/SAVE(1)[0]:The user chose Y when decidin
g whether to save the configuration to the device.
Save the configuration successfully.

```

```

R2
The device is running!

<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>sys
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2]dhcp enable
Info: The operation may take a few seconds. Please wait for a moment.done.
[R2]interface Eth
[R2]interface Ethernet 0/0/0
[R2-Ethernet0/0/0]ip address 150.100.12.2 24
      ^
Error: Wrong parameter found at '^' position.
[R2-Ethernet0/0/0]ip address 150.100.12.2 24
[R2-Ethernet0/0/0]q
[R2]interface Eth
[R2]interface Ethernet 0/0/1
[R2-Ethernet0/0/1]ip address 172.16.1.1 24
[R2-Ethernet0/0/1]dhcp select relay
[R2-Ethernet0/0/1]dhcp relay server-ip 150.100.12.1
[R2-Ethernet0/0/1]q
[R2]q
<R2>save
The current configuration will be written to the device.
Are you sure to continue?[Y/N]Y
Info: Please input the file name ( *.cfg, *.zip ) [vrpcfg.zip]:
Now saving the current configuration to the slot 17.
Save the configuration successfully.

```

For 3.3.3.0, 1.1.1.0 network pings

```

<R2>u t m
Info: Current terminal monitor is off.
<R2>sys
<R2>system-view
Enter system view, return user view with Ctrl+Z.
[R2]ip route-static 172.168.1.0 24 150.100.12.1
[R2]q
<R2>sys
Enter system view, return user view with Ctrl+Z.
[R2]undo ip route-static 172.168.1.0 24 150.100.12.1
[R2]ip route-static 0.0.0.0 0 150.100.12.1
[R2]q
<R2>sys
Enter system view, return user view with Ctrl+Z.
[R2]q
<R2>save
The current configuration will be written to the device.
Are you sure to continue?[Y/N]Y
Now saving the current configuration to the slot 17.
Save the configuration successfully.

```

Feature	global	interface
Uses DHCP pool	Manual (you create it)	Auto (from interface IP)
Flexibility	High (DNS, exclusions, options)	Limited
Best for	Larger networks	Small/simple setups
Config needed	Yes (ip pool)	No

Use DHCP relay when, Clients and DHCP server are in **different subnets** AND You want centralized DHCP server

Example:

- PCs → 172.16.1.0/24
- DHCP Server → 150.100.12.1 (different network)

Relay makes them communicate

```

PC6
Basic Config Command MCPacket UdpPacket Console
Physical address.....: 54-89-98-EA-34-75
DNS server.....:

PC>ipconfig /renew

IP Configuration

Link local IPv6 address.....: fe80::5689:98ff:feea:3475
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 172.16.1.248
Subnet mask.....: 255.255.255.0
Gateway.....: 172.16.1.1
Physical address.....: 54-89-98-EA-34-75
DNS server.....:

PC>ping 1.1.1.1

Ping 1.1.1.1: 32 data bytes, Press Ctrl_C to break
Request timeout!
Request timeout!
Request timeout!
Request timeout!
Request timeout!

--- 1.1.1.1 ping statistics ---
    5 packet(s) transmitted
    0 packet(s) received
   100.00% packet loss

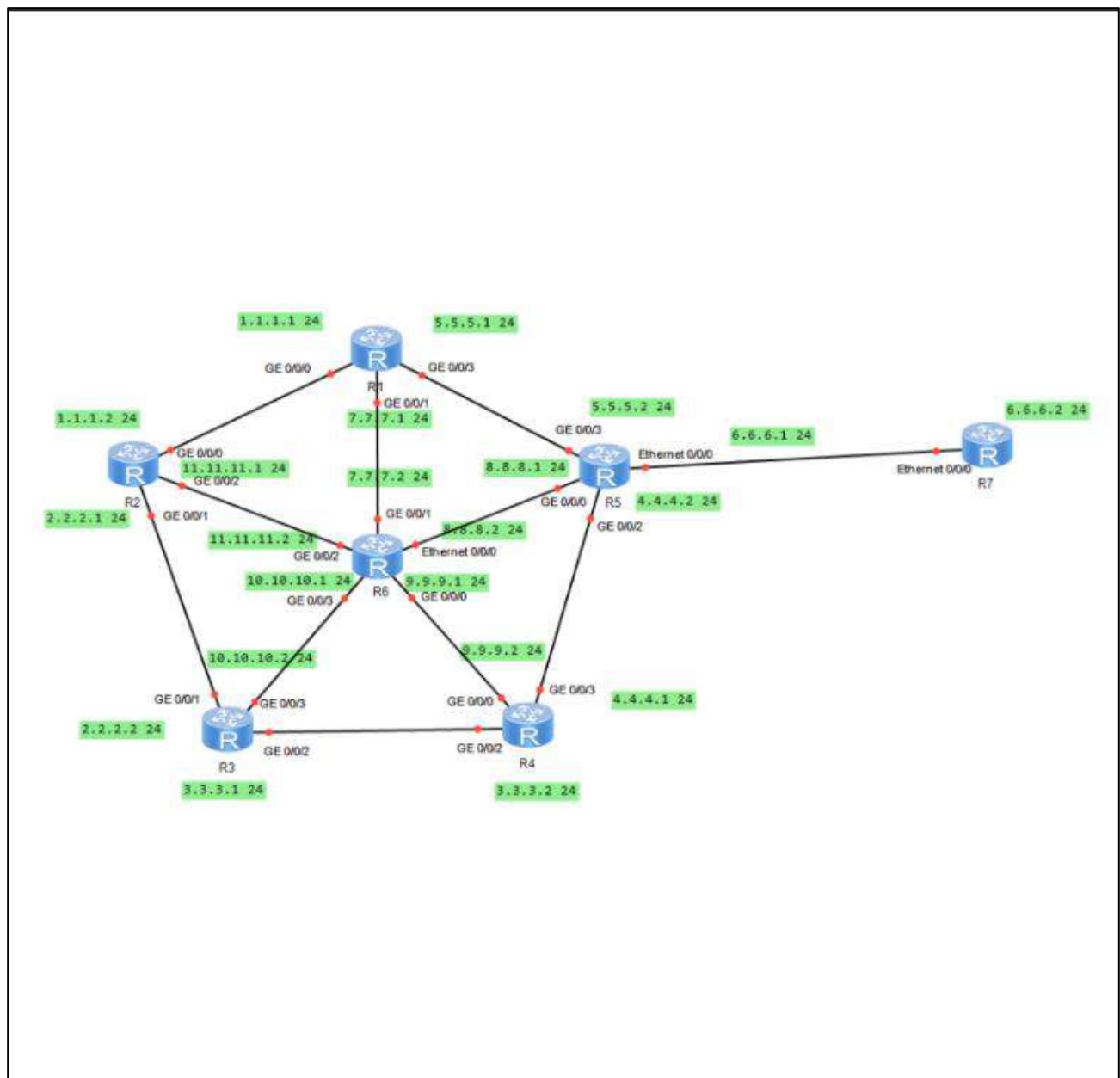
PC>ping 1.1.1.1

Ping 1.1.1.1: 32 data bytes, Press Ctrl_C to break
From 1.1.1.1: bytes=32 seq=1 ttl=254 time=78 ms
From 1.1.1.1: bytes=32 seq=2 ttl=254 time=78 ms
From 1.1.1.1: bytes=32 seq=3 ttl=254 time=63 ms
From 1.1.1.1: bytes=32 seq=4 ttl=254 time=63 ms
From 1.1.1.1: bytes=32 seq=5 ttl=254 time=94 ms
  
```

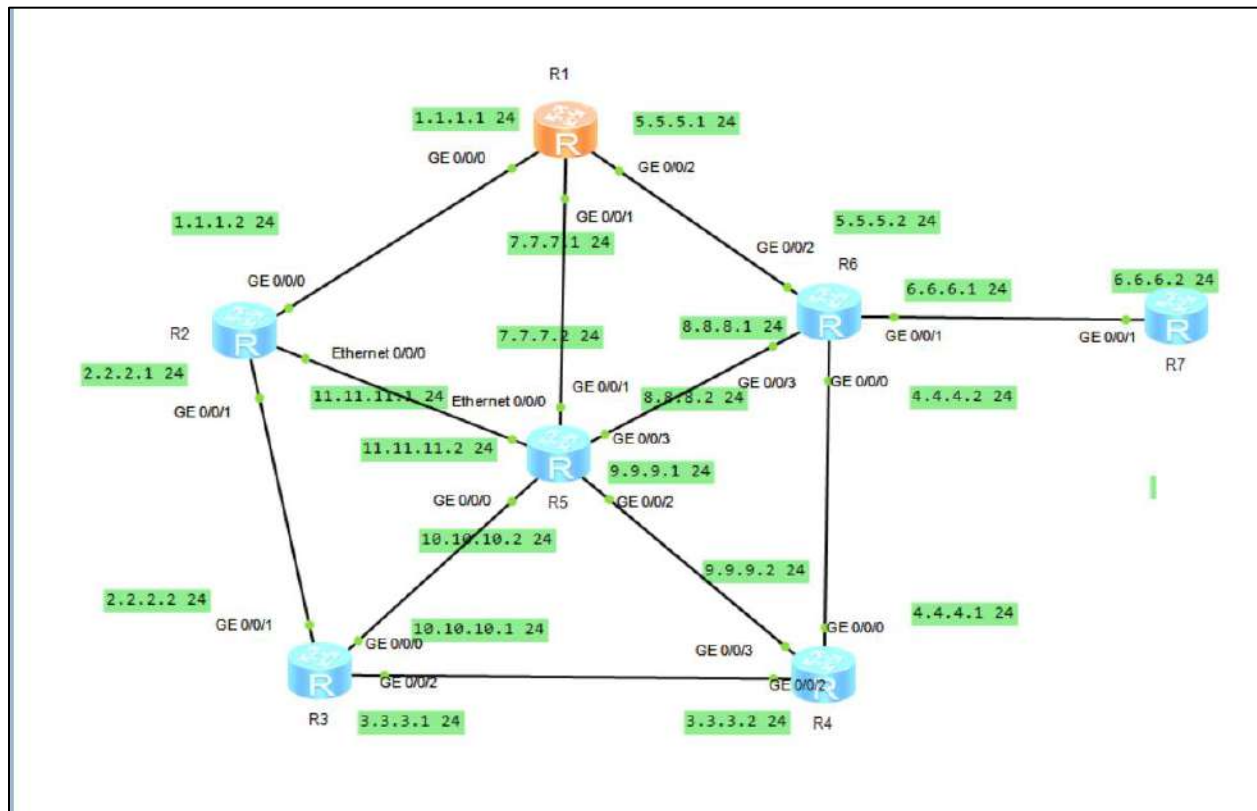
LAB 12

Exercise:

Implement the following scenario by applying RIP protocol (version 2).



Implementation:



```

R1
The device is running!

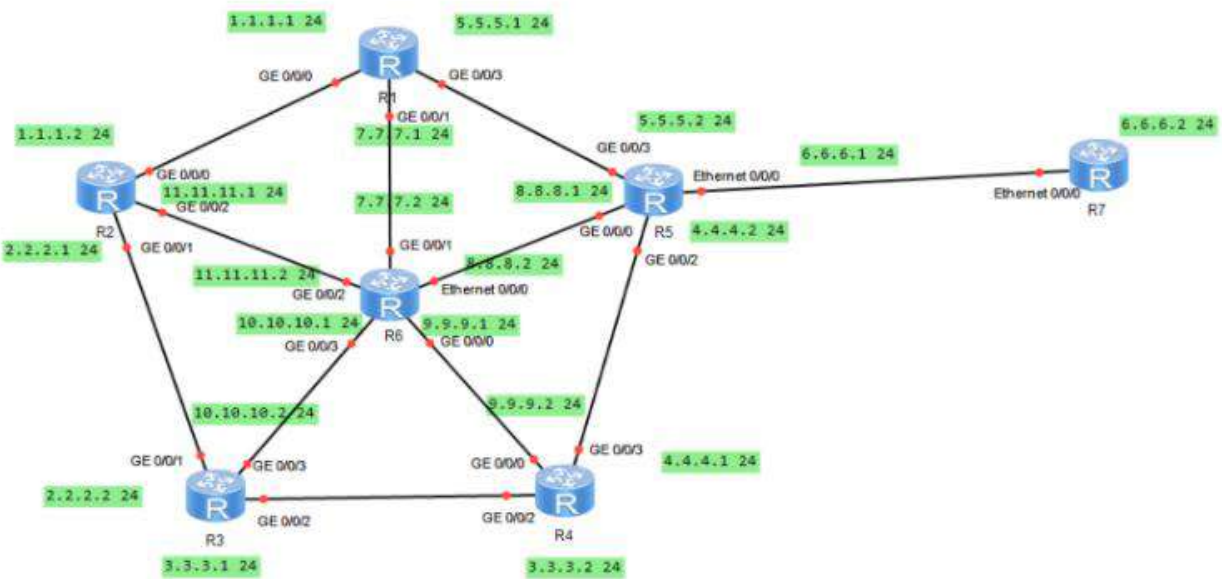
<Huawei>ping 4.4.4.1
  PING 4.4.4.1: 56 data bytes, press CTRL_C to break
    Reply from 4.4.4.1: bytes=56 Sequence=1 ttl=254 time=150 ms
    Reply from 4.4.4.1: bytes=56 Sequence=2 ttl=254 time=40 ms
    Reply from 4.4.4.1: bytes=56 Sequence=3 ttl=254 time=130 ms
    Reply from 4.4.4.1: bytes=56 Sequence=4 ttl=254 time=70 ms
    Reply from 4.4.4.1: bytes=56 Sequence=5 ttl=254 time=70 ms

  --- 4.4.4.1 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 40/92/150 ms
  
```

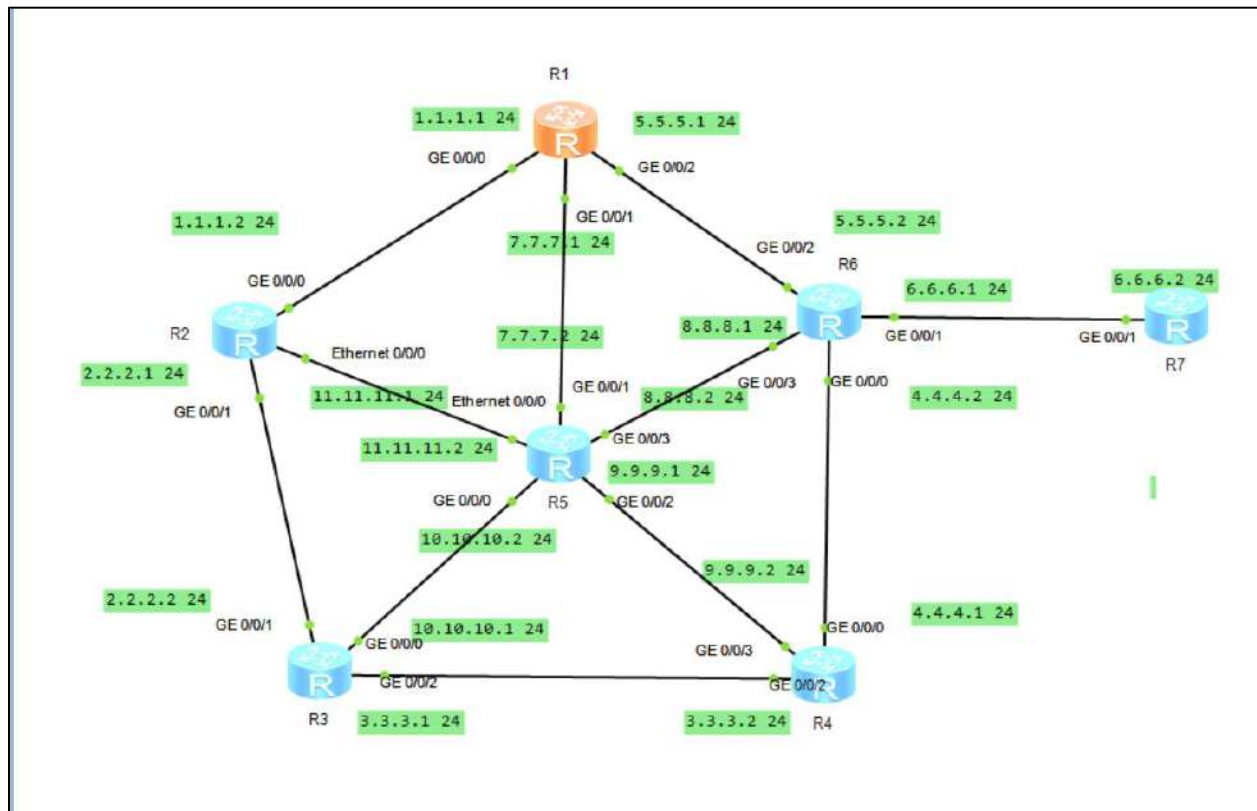
LAB 13

Exercise:

Add console and vty security access settings in each router of Lab 12.



Implementation:

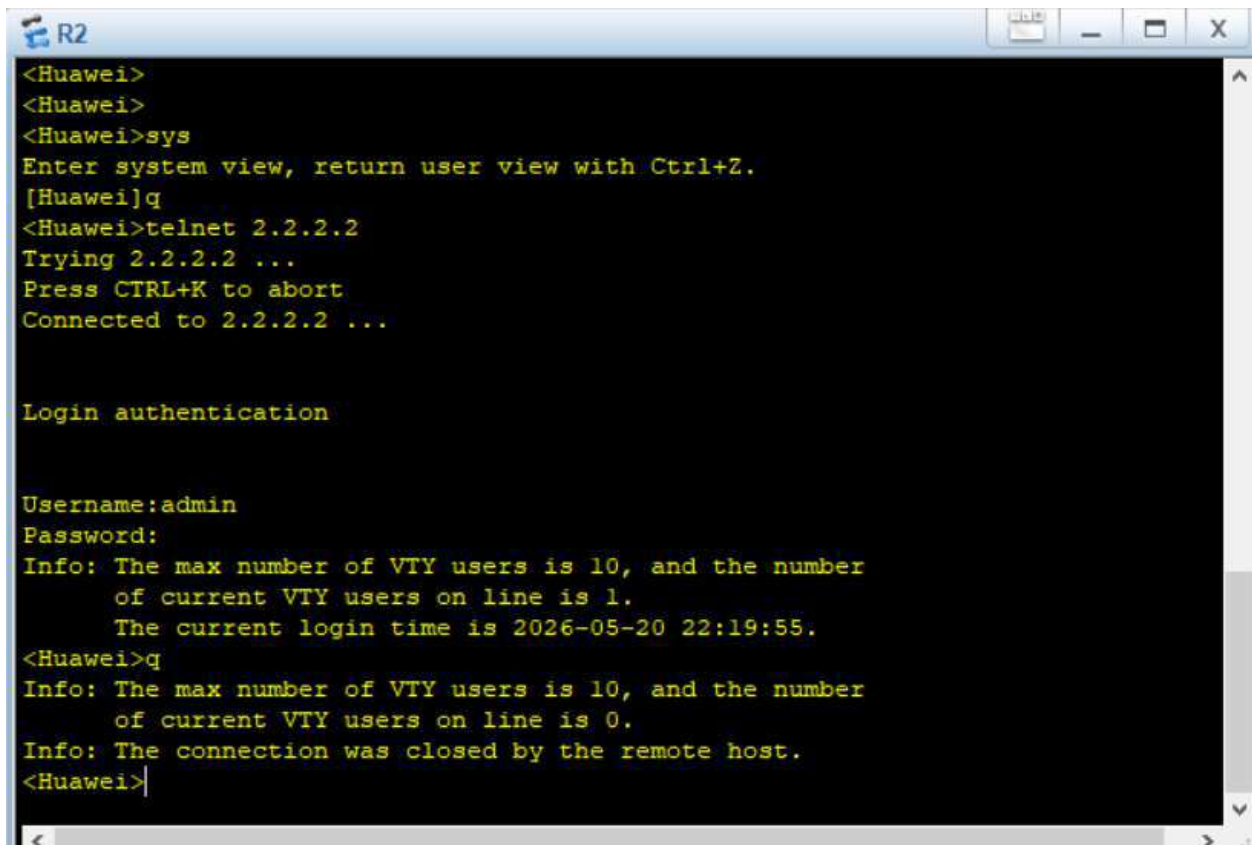


```

The device is running!

<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>sys
Enter system view, return user view with Ctrl+Z.
[Huawei]telnet server enable
Info: The Telnet server has been enabled.
[Huawei]user-interface vty 0 4
[Huawei-ui-vty0-4]authentication-mode aaa
[Huawei-ui-vty0-4]protocol inbound telnet
[Huawei-ui-vty0-4]q
[Huawei]aaa
[Huawei-aaa]local-user admin password cipher admin123
[Huawei-aaa]local-user admin service-type telnet
[Huawei-aaa]local-user admin privilege level 3
[Huawei-aaa]q
[Huawei]

```



```
<Huawei>
<Huawei>
<Huawei>sys
Enter system view, return user view with Ctrl+Z.
[Huawei]q
<Huawei>telnet 2.2.2.2
Trying 2.2.2.2 ...
Press CTRL+K to abort
Connected to 2.2.2.2 ...

Login authentication

Username:admin
Password:
Info: The max number of VTY users is 10, and the number
      of current VTY users on line is 1.
      The current login time is 2026-05-20 22:19:55.
<Huawei>q
Info: The max number of VTY users is 10, and the number
      of current VTY users on line is 0.
Info: The connection was closed by the remote host.
<Huawei>
```